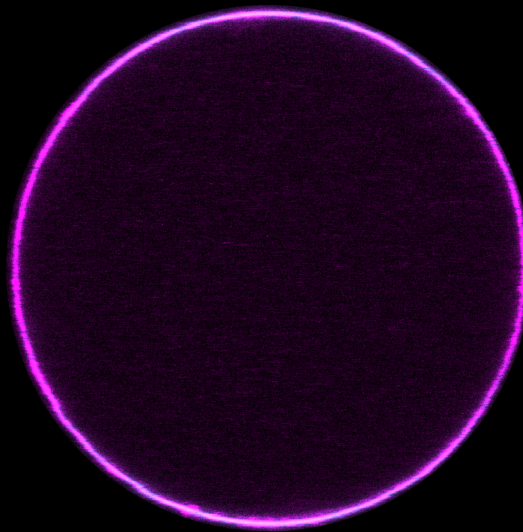


**Reconstitution of a septin cortex inside
Giant Unilamellar Vesicles:**

*Quantification of septin localization and its effect
on membrane mechanics*



Master Thesis Project

Aikaterini-Maria Kourkoulou

April 2023



Master of Applied Physics

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by

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Abstract

Septins are a family of conserved eukaryotic proteins that has recently been acknowledged as the fourth component of the cytoskeleton. They can assemble into palindromic hetero-oligomeric complexes that polymerize, forming higher order structures such as filaments, bundles, and meshes. As suggested in literature, since membrane-associated meshwork formation has been observed in absence of other cytoskeletal components, a septin cortical network could actively participate in the cell cortex, directly affecting cell surface mechanics. To investigate this possibility, we encapsulate human hexameric and octameric complexes in Giant Unilamellar Vesicles (GUVs) using emulsion Droplet Interface Crossing Encapsulation (eDICE) and quantify septin localization on the membrane for different membrane compositions. We then assess the effect of septin-membrane interactions on membrane mechanics using Vesicle Fluctuation Analysis (VFA). We show that human septins bind to negatively charged membranes containing DOPS or PIP2, but not to neutral DOPC membranes. We identify a synergistic effect in septin recruitment for DOPS and PIP2 and observe that septin localization on the membrane is highly sensitive to the presence of PIP2. Finally, we encounter a subset of vesicles with higher bending rigidities in samples with GUV membranes containing DOPS and PIP2, that is absent in samples of neutral DOPC membranes, indicating that septin binding can contribute in membrane rigidification.

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List of Abbreviations

GUV	Giant Unilamellar Vesicle
LUV	Large Unilamellar Vesicle
SUV	Small Unilamellar Vesicle
SLB	Supported Lipid Bilayer
PC	Phosphatidylcholine
DOPC	1,2-dioleoyl-sn-glycero-3-phosphocholine
POPC	1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine
PS	Phosphatidylserine
DOPS	1,2-dioleoyl-sn-glycero-3-phospho-L-serine
PI	Phosphatidylinositol
PIP2	Phosphatidylinositol-4,5-bisphosphate
Biotin-PE	1-oleoyl-2-(12-biotinyl (aminododecanoyl))-sn-glycero-3-phosphoethanolamine
Rhodamine-PE	1,2-dioleoyl-sn-glycero-3-phosphoethanolamine-N-(lissamine rhodamine B sulfonyl)
msfGFP	monomeric superfolder Green Fluorescent Protein
GTP	Guanosine Triphosphate
MgATP	Magnesium-Adenosine Triphosphate
SPB	Septin Polymerization Buffer
IAS	Inner Aqueous Solution
OAS	Outer Aqueous Solution
VB	Vesicle Buffer
eDICE	emulsion Droplet Interface Crossing Encapsulation
cDICE	continuous Droplet Interface Crossing Encapsulation
VFA	Vesicle Fluctuation Analysis
QCM-D	Quartz Crystal Microbalance with Dissipation monitoring
AFM	Atomic Force Microscopy
FRAP	Fluorescence Recovery After Photobleaching
TEM	Transmission Electron Microscopy
SEM	Scanning Electron Microscopy

1. Introduction

1.1. The mammalian plasma membrane

1.1.1. Function and composition

The mammalian plasma membrane plays a crucial role in the sustenance and proliferation of the cell. Its function is not only limited to passive compartmentalization, with the separation of the cytosol from the extracellular environment, but also involves essential biological dynamic processes [1, 2]. Through interactions with other cellular elements, such as the cytoskeleton [3, 4], a complex dynamic network of protein filaments present in the cytoplasm, membrane-shape remodelling can directly and closely regulate cell functions [5], accommodate for the transportation of substances in or out of the cell [6], and provide a substrate for surface reactions in signaling [7].

This diverse range of activities is supported by a heterogeneous structure that can undergo dynamic rearrangements [8]. The basis of the plasma membrane is a bilayer of phospholipids. Phospholipids consist of a polar hydrophilic headgroup and two hydrophobic hydrocarbon chains, commonly referred to as tails. Due to their amphiphilic nature, in presence of aqueous solutions phospholipids spontaneously self-assemble to minimize the hydrophobic surface area that is exposed to water. To this end, phospholipids can assemble into spherical structures with the tails facing inwards, called micelles, or double layers, with their heads exposed to water on both sides and their tails enclosed in the interior forming a hydrophobic core [9] (Fig. 1.1A).

In cells, an enormous amount of different phospholipid species is encountered, with variances both in the polar headgroups as well as in the length and saturation of the acyl chains. The phospholipid species present and their prevalence in the plasma membrane vary per cell type and also differ between the inner and the outer leaflet of the lipid bilayer [10]. One of the most abundant phospholipids types in mammalian cell membranes is Phosphatidylcholine (PC) [11]. PC molecules are overall neutral. In contrast, other phospholipid molecules encountered in plasma membranes are negatively charged, as in the case of Phosphatidylserine (PS) and Phosphatidylinositol (PI), with net charges of -1 and up to -4 respectively [12]. Those phospholipids are mainly concentrated in the inner leaflet of the plasma membrane and participate in protein binding and membrane dynamics [13, 14].

On the approximately 6 nm thick lipid bilayer [2] that forms the basis of the plasma membrane other components such as sterols and proteins are embedded (Fig. 1.1B), that aid in both the membrane's structure and function. The protein content of the plasma membrane is high, comprising approximately half of the total membrane mass [8]. Proteins can be integrated in the lipid bilayer or be bound to it through specific protein domains or through electrostatic interactions with lipid headgroups [8]. The protein-membrane affinity can be highly specific to certain phospholipid species, allowing for controlled spatial protein distribution and regulation of various cellular functions [13, 15].

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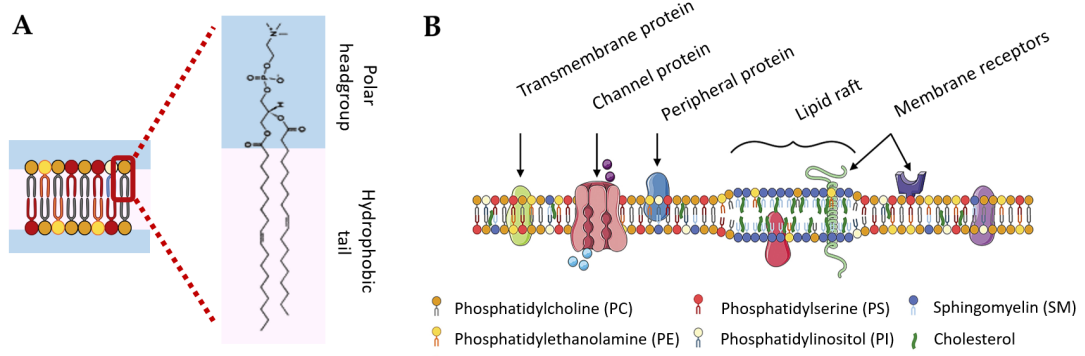


Figure 1.1.: **Schematic representation of the structure of the mammalian plasma membrane.** (A) The basis of the membrane is a lipid bilayer formed by a plethora of different phospholipids. Due to their amphipathic nature, in aqueous solutions phospholipids can assemble in a bilayer, forming a hydrophobic core. The molecular structure of DOPC is depicted on the right, indicating the hydrophilic headgroups and the hydrophobic tail. (B) The membrane is a complex system, containing various lipids and a variety of embedded proteins. Figures are adapted from [2].

1.1.2. Membrane mechanics

The importance of lipid membranes in cell shape control and mechanical integrity has generated great interest in studying membrane mechanics. The most broadly used mathematical description of biological membranes is a continuum mechanical model introduced in 1973 by W. Helfrich [16]. As biological membranes are very thin, they are considered as two-dimensional surfaces existing in a three-dimensional space. Membrane shape at each point in space can be characterized by two curvatures, called the principal curvatures, c_1 and c_2 . Those curvatures correspond to the maximum and minimum values of curvature for all normal planes drawn at that point (Fig. 1.2A). Two combinations of them are mostly used, called the mean (H) and the Gaussian (K) curvature [17]:

$$H = \frac{1}{c_1} + \frac{1}{c_2} \quad \text{and} \quad K = c_1 c_2 \quad (1.1)$$

The contribution of the membrane curvature to the total free energy of the membrane is described in the Helfrich-Canham Hamiltonian [16, 18], that relates the energy density to the mean and Gaussian curvatures, H and K , the spontaneous curvature, C_0 , and the membrane tension, σ , with an integration over the membrane area, S :

$$F_{\text{Helfrich}} = \int_S \left[\frac{\kappa}{2} (2H - C_0)^2 + \bar{\kappa} K + \sigma \right] dA \quad (1.2)$$

The parameters κ and $\bar{\kappa}$ are called the bending and Gaussian elastic modulus, respectively. In the case of closed membranes that have no holes, such as spheres, according to the Gauss-Bonnet theorem [19], the Gaussian curvature term gives a constant contribution and can thus

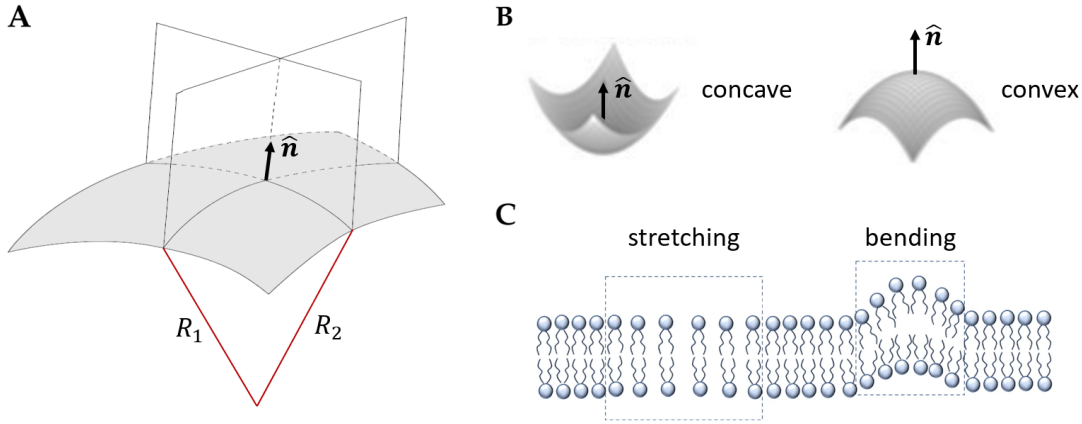


Figure 1.2.: Membrane curvature and deformations. (A) Biological membranes can be described as a 2D surface in 3D space. To measure the curvature of the surface at a specific point in space we can draw the normal vector, $\hat{n}$, and look at the cross-section of a plane that contains $\hat{n}$ and the membrane surface (grey highlighted area). The curvature of the cross-section can be defined by the radius (marked with red) of the corresponding osculating circle and is generally different for different selected normal planes. Membrane shapes at each point are characterized by the two principal curvatures which correspond to the maximum and minimum values of curvature for all normal planes, $c_1=1/R_1$ and $c_2=1/R_2$. (B) Curvature can be either negative or positive, corresponding to concave and convex areas, respectively. (C) External forces can cause the membrane to stretch or bend. Stretching of a lipid bilayer increases the hydrophobic area exposed to water, leading to an energy cost. Bending deformations require stretching and compression of each membrane leaflet, increasing the energy of the system. Figure adapted from [21].

be neglected, limiting the interest in $\bar{\kappa}$. When in addition no asymmetries exist between the two leaflets of the membrane bilayer, for example, in absence of lipid composition differences, the spontaneous curvature C_0 equals zero and then Eq. (1.2) is simplified to:

$$F_{Helfrich} = \int_S \left[\frac{\kappa}{2} (2H)^2 + \sigma \right] dA \quad (1.3)$$

Eq. (1.3) indicates two main parameters that influence the membranes elastic energy, the membrane tension, σ , and the bending rigidity, κ . The membrane tension relates to forces exerted in a tangential plane to the membrane's surface, indicating resistance to stretching deformations (Fig. 1.2C, left). On the other hand, the bending modulus, often referred to as bending rigidity, is a quantification of the energy cost required for the compression and expansion of the two membrane leaflets when changes of curvature are induced due to forces in the normal direction of the membrane surface (Fig. 1.2C, right). Therefore, κ constitutes an important measure for the assessment of the membrane's flexibility. The determination of these membrane elastic properties can provide insight into morphological transitions linked to cellular processes or highlight the effect of interactions with other cellular components, such as a membrane-associated meshwork of protein filaments. Thus, various experimental approaches have been employed in order to determine the membrane elastic properties, with a special focus on the bending rigidity [20].

1.2. The cell cortex

Plasma membrane function is closely tied to its geometry. In order to accommodate for vital cellular functions, such as cell division, migration, and morphogenesis, the membrane needs to be deformable and appropriately change its shape [5]. At the same time, the presence of external mechanical forces creates the requirement for a rigid container for the protection of the cell [22]. However, when overly rigid, cells are brittle and prone to fracture due to stress, while, when too flexible, cells cannot generate the forces required to maintain or alter their shape [4]. Therefore, cells need a balance between rigidity and elasticity and the ability to dynamically change the membrane topography in a precisely controlled way.

As the plasma membrane is very thin and fragile there is the need for a supportive network to exist to regulate the cellular shape and the membrane mechanics [3]. In the case of animal cells [23], this role is carried out by the cytoskeleton, a dynamic filamentous protein network present in the cytoplasm. The cytoskeleton has been thought to mainly consist of three filament types, namely actin filaments, microtubules, and intermediate filaments. Each type has its own characteristic structure and cellular distribution and results from the polymerization of distinct types of protein subunits. Cytoskeletal elements interact extensively with cellular membranes and each other and play a vital role in regulating cell division, polarity, and cell stiffness [24].

In particular, membrane structure scaffolding is mainly accomplished by a cross-linked actin filament network that lies directly underneath the plasma membrane, known as the cell cortex [25]. The dense crosslinked actin meshwork is accompanied by a great variety of actin-binding proteins that participate in actin assembly and disassembly, actin cross-linking, actin-membrane anchoring, and force and movement generation [26]. With their assistance, the actin filaments can facilitate the generation of contractile forces on the cortical network, ultimately controlling membrane surface tension. Modifications in the cell cortex composition and organization translate into tension gradients, leading to membrane shape deformations and contractions required in vital cell functions such as cell migration and division [5, 26].

In a recent study by Laplaud et al [27] using dendritic cells from mice, the cellular cortex was pinched between two beads with the use of an external magnetic field allowing the measurement of its mechanical properties. Interestingly, when actin polymerization was impeded and the actin filaments of the cytoskeleton were disassembled with the addition of a toxin, the measured stiffness and thickness remained higher than what is expected for a naked membrane. This is an indication that other proteins present in the cell cortex might have a more direct influence on membrane mechanics than mere indirect control through actin organization and actin membrane-anchoring regulation. One of the possible cortical elements to have such a contribution is septin, a family of proteins known to interact with rigidity-regulating systems, as actin and microtubules [4], and to assist in actin-membrane binding [28]. Due to their association with membranes and other cortical elements and their ability to form filaments, septins have been recently acknowledged as a fourth component of the cytoskeleton, although they have been less investigated than the other cytoskeletal components [28].

1.3. Human septins

1.3.1. Structure and polymerization

Septins form a family of GTP-binding proteins expressed in eukaryotic cells. The number of septin encoding genes in an organism varies widely between different species. In the case of humans, there are 13 different septin genes, SEPT1-SEPT12 and SEPT-14, that are classified in four groups based on sequence similarities, as depicted in Fig. 1.3A. All human septin subunits have a central core domain consisting of a polybasic region near the N-terminal, a GTP-binding domain, and a highly conserved sequence of 53 amino acids of undetermined function, known as the Septin Unique Element (SUE). At the N- and C-termini there are respectively a Pro-rich region and a Coiled-coil domain, which vary in composition and length. The function of the different domains is not conclusively determined [28].

In their folded conformation, different septin subunits can interact with each other through two interfaces, the G-interface that is comprised of the GTP-binding domain and the NC-interface formed by the N and C-termini [29]. The septin subunits assemble by alternating G and NC-interfaces, forming linear palindromic hetero-oligomeric complexes (Fig. 1.3B, and Fig. 1.4). Human septins form hetero-hexamers and heter-octamers with the following subunit arrangements: SEPT2-SEPT6-SEPT7-SEPT7-SEPT6-SEPT2 [30] and SEPT2-SEPT6-SEPT7-SEPT3-SEPT3-SEPT7-SEPT6-SEPT2 [31], with lengths of 24 nm [32] and 32 nm [33] respectively. From the SEPT3 subfamily, SEPT9 is the most widespread and most typically encountered in cellular septin octameric complexes [34, 35]. Septin hetero-oligomers can in turn polymerize into non-polar filaments, by joining end-to-end through their SEPT2 subunits [29]. Different filaments can then associate laterally and form higher order structures such as paired filaments, bundles, rings, cage-like formations, and meshes [28, 33, 36] (Fig. 1.4). Septin polymerization in a solution can be inhibited (~300 mM) or promoted (<150 mM) by variation of the salt concentration [37].

1.3.2. Function in the cell and the cytoskeleton

In the form of higher order structures, septins contribute to a plethora of cellular functions. Originally discovered during research on budding yeast [39], septins play an essential role in cytokinesis and bud growth [40]. They are encountered at the bud neck of yeast [41], as well as at the cell division site of mammalian cells [42], arranged in two rings aligned circumferentially around the bud neck [41] that regulate the localization and accumulation of other proteins necessary for cell division [43].

Apart from dividing cells, septins are also found on the plasma membrane, acting as membrane scaffolds of membrane-bound proteins and surface receptors to control their distribution in the cell, possibly through further interactions with the cortical actin filaments [28]. They additionally serve as diffusion barriers, usually in the form of rings, compartmentalizing and isolating membrane proteins to specific cellular domains, as in the case of the annulus of the mammalian spermatozoa [44] and the base of cilium [45], where they are required for both structural and mechanical integrity. The diversity in septin activity is further highlighted by its participation during bacterial infection where septin, as other cytoskeletal elements, undergoes rearrangements forming ring-base cages around invading bacteria, restricting their motility and assisting in bacterial autophagy [46].

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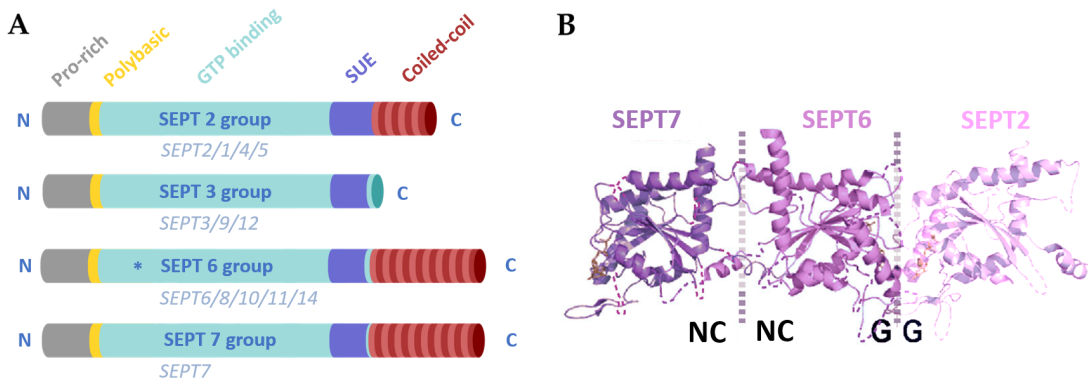


Figure 1.3.: Septin structure and subunit interfaces. (A) The 13 human septin genes are categorized in four groups (SEPT2, SEPT3, SEPT6, SEPT7) based on sequence similarities. The members of each group are indicated in italics. All subunits have three conserved domains, the polybasic region close to the N-terminus, the GTP-binding domain and the septin unique element (SUE). A Pro-rich and a Coiled-coil domain are present at the N and C- termini, respectively. Notably, the SEPT3 group does not include a Coiled coil domain. Septins belonging to the SEPT6 group lack a Thr residue (Thr78), which prevents them from hydrolysing GTP to GDP, indicated by an asterisk. Figure adapted from [28]. (B) X-ray crystal structure of SEPT2/6/7. In their folded state, septin subunits interact through NC-NC and G-G interfaces, forming linear palindromic hetero-oligomers. Figure adapted from [38].

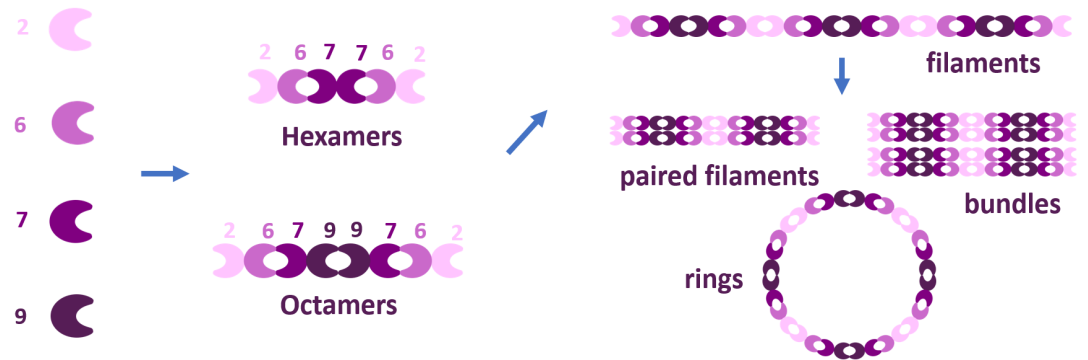


Figure 1.4.: Septin assembles into higher order structures. Human septin subunits can form hetero-hexamers and hetero-octamers, which in turn can polymerize into filaments that can assemble into higher ordered structures, such as paired filaments, bundles and rings.

Growing evidence indicates that septin assemblies play an essential role in the stabilization of membranes and cellular rigidity. Septins directly bind with lipid membranes [36, 41, 47] and can interact with actin, microtubules, and myosin-II, a motor protein responsible for the generation of large contractile forces in the cell cortex, affecting their activity and moderating their interactions with the plasma membrane [4]. Septin depletion has been demonstrated to lead to mechanically impaired and deformed cells. In the case of T cells, removal of septin complexes led to unexpected blebbing, membrane elongations and problematic motility, a result one would expect from the disturbance of the cellular cortical network [48]. Similarly, in the aforementioned spermatozoa annulus, septin depletion seems to again impair motility as much as structural integrity [44]. Combining these results with Laplaud's observation discussed in Section 1.2, it is highly suggestive that a septin filament meshwork could provide direct cortical rigidification even in absence of an actin network.

1.4. Bottom-up synthetic biology

Due to the great complexity of the composition and the interconnected interactions of the different cellular components, it is challenging to decode the mechanical and biochemical mechanisms present in the cell. To obtain an improved understanding of cellular components and functions, a commonly adopted approach is the study of isolated components of interest in a controlled environment. This approach is called bottom-up synthetic biology [49]. By purification and recombination of selected cellular components in reconstituted, cell-mimicking minimal systems, we can gain quantitative insights in the contribution of individual cellular elements in different cellular processes. Combining the obtained knowledge, more complex reconstituted systems can be gradually constructed, simulating different cellular mechanisms and expanding our biophysical knowledge [50].

1.4.1. Mimicking the plasma membrane

The inherent complexity and diversity of the plasma membrane pose a challenge in studying its biophysical properties and understanding the effect of individual proteins, such as septins, on the membrane mechanics. Following the bottom-up approach, it is of interest to construct a minimal membrane model system that still maintains some characteristics of its biological counterpart and can thus provide useful biologically relevant conclusions [51].

In a 2D geometry, lipid monolayers (Fig. 1.5A), consisting of a few selected phospholipids and lacking any embedded elements, can be used for studies of membrane-protein interactions [52]. Similarly, supported lipid bilayers (SLBs) that are spread over a solid support, such as glass, can be employed [2] (Fig. 1.5B). Such systems allow for precise control of the chemical environment and for surface-sensitive high-resolution measurement techniques. Additionally, curvature can be induced with the use of an undulating supporting surface template or through SLB coating of beads to mimic 3D cell-like geometry [33, 41].

However, the presence of a solid support poses an obstacle in probing the mechanical properties of membranes, where free-floating membrane systems are required. For the purpose of such studies, Giant Unilamellar Vesicles (GUVs) are often chosen as a minimal membrane model [51]. Unilamellar vesicles are spherical single lipid bilayers that encapsulate and separate an aqueous solution from the surrounding aqueous medium (Fig. 1.5C). GUVs assemble

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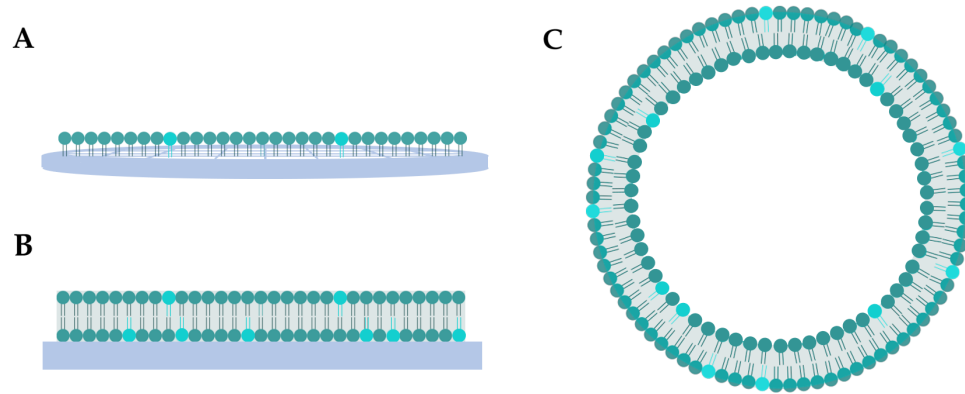


Figure 1.5.: **Schematic representation of model membrane systems used in protein studies.** (A) Lipid monolayers are often prepared on a grid support for transmission electron microscopy (TEM) experiments. (B) Supported lipid bilayers (SLBs) are formed after fusion of SUVs on a solid substrate and can similarly be employed for high-resolution measurement techniques in studies of protein-membrane interactions. (C) Free-floating Giant Unilamellar Vesicles (GUVs) can encapsulate a different aqueous solution than the outer aqueous solution and allow for probing the mechanical properties of membranes.

due to the energetic cost of the border regions of flat lipid bilayers. These containers are categorized as Small, Large, or Giant Unilamellar vesicles (SUVs, LUVs and GUVs) based on their size (smaller than 100nm, 100nm-1 μ m and larger than 1 μ m respectively) [2]. As GUVs are comparable in size to eukaryotic cells, they provide a well-controlled and relevant membrane model system. As some production protocols also allow for the encapsulation of other biomolecules such as proteins [53, 54, 55, 56, 57], GUVs are a very convenient system for the study of biologically relevant membrane mechanics.

1.5. In vitro septin-membrane studies

1.5.1. Septin localization on the membrane

Phosphatidylinositol 4,5-bisphosphate (PI(4,5)P₂ or PIP₂), albeit constituting only about 1% of the lipids in the plasma membrane, is one of the most important phospholipids in the regulation of a variety of cellular functions [12]. With a net charge of -4 at pH 7.0, through interactions mediated by the headgroups, PIP₂ is known to specifically interact with various proteins, anchoring them to the plasma membrane and determining their spatial distribution on the membrane surface [58]. PIP₂ is involved in actin cytoskeleton dynamics and is also known to appear enriched during mitosis in the bud neck of budding yeast [59].

Previous research work involving reconstituted yeast septin complexes and lipid monolayers [52] or bilayers [60, 61, 62] has demonstrated that septin-membrane binding shows a high selectivity for PIP₂, as well as other phosphatidylinositol molecules such as PI(4)P, PI(5)P and PI [61, 63, 64, 65, 66]. However, in contrast to yeast septin, a recent study with fly septin hexamers by Szuba et al. [36] has reported similar lipid affinity for PS as well as PIP₂. Specifically, preparation of SLBs with a content of 5%-20% PS showed equivalent septin

recruitment on the membrane as in the case of membranes with 1%-8% PIP2. The different concentration thresholds for septin organization for the two lipids were consistent with the difference of the respective net charges. As, additionally, fly septins did not adhere to neutral 100% DOPC membranes, there are strong indications that electrostatic forces are important in their interactions with membranes. In the same study, it was demonstrated that septin-membrane binding can promote septin polymerization and ordered self-assembly, possibly due to 2D confinement and/or controlled septin re-orientation.

1.5.2. Membrane-associated septin meshworks

Fly septins *in vitro* have been observed to form membrane-bound filamentous meshworks even in absence of other cytoskeletal elements [36]. On 20% PS plus 5% PIP2 SLBs and in concentrations larger than 200 nM, fly septins formed a filamentous network comprised of paired filaments bridged by isolated hexamer rods. Atomic force microscopy (AFM) experiments revealed that an increase of septin concentration causes the paired filaments to bundle, leading to a dense septin network that includes septin stacking through the form of a second septin layer. The observed septin meshwork appeared to be stably anchored to the membrane, displaying minimal subunit exchange after fluorescence recovery after photobleaching (FRAP), whereas the lipid bilayer underneath displayed fluid behavior. In agreement, repeated AFM scans indicated that bundles can be laterally displaced along the membrane, without being destroyed, showcasing a mechanically stiff septin meshwork with a fluid link to the membrane.

The formation of a membrane-bound septin meshwork has also been observed for human septins. In our group, in a study carried out by Djim de Ridder with transmission electron microscopy (TEM) on lipid monolayers containing 5% PIP2, incubation with septin hexamers led to the formation of a dense cross-hatched structured meshwork. The meshwork displayed arrays of paired filaments perpendicular to parallel aligned single filaments. The spacing between paired filaments was regular and comparable to the length of a hexamer (24nm) [32]. Similarly, Nakazawa et al. [33] using scanning electron microscopy (SEM) demonstrated that human septin octamers assemble in a two-layered orthogonal filament network on undulated supported lipid bilayers containing 10% DOPS and 8% PIP2. This septin network fully saturated for septin concentrations above 90 nM, was uniform for both convex and concave regions and showed a regular inter-filament spacing of approximately 30 nm, similar to the octameric length (32 nm). In our group, FRAP experiments conducted by Gerard Castro-Linares on SLBs containing 5% PIP2 incubated with either hexamers or octamers showed no signs of septin subunit exchange, indicating that the septin meshes are stably anchored on the membrane, similarly as in the case of fly septins [36].

1.5.3. Curvature dependence of septin affinity

Septins are often encountered in plasma membrane regions that present a micro-scale curvature, as in the case of the division site [42] and the base of cilia [45]. Indeed, previous studies on yeast and human septins have demonstrated that septin filaments are able to sense [41, 67] and also induce curvature in the micron-scale [33, 41] .

With the use of silica beads coated with lipid bilayers containing 25% PI phospholipids, Bridges et al.[67] showed that yeast and human septin complexes in physiological conditions recognise positive curvature and preferentially bind to curvatures of $2 \mu\text{m}^{-1}$ from a

1. Introduction

tested range of $0.31\text{--}6.67\ \mu\text{m}^{-1}$. This behaviour was also maintained in the case of high salt concentrations that prevented polymerization, demonstrating that single septin complexes are also able to detect curvature variances in the micro-scale.

At the same time, septins also seem to differentiate between positive and negative curvatures. With the use of solid periodically wavy templates that were coated with SLBs containing PIP₂, incubation with both yeast [41] and human [33] septins indicated that septins preferentially bind to negatively curved regions at low concentration ($<26\ \text{nM}$). With further increase of septin concentration, septin filaments could be located throughout the substrate and were observed to align themselves in a flat manner parallel to the null-curvature axis of positively curved hills but bent along following the negative curvature of the substrate's valleys. Septins' avoidance to being positively curved was further demonstrated when they were added in the exterior of free-floating and slightly deflated GUVs. Both yeast [41] and human [33] septins were reported to deform the membrane upon localization, inducing periodic bump-like deformations with micro-scale negatively curved areas between the deformations' peaks. These findings indicate that septins can actively generate their preferable curvatures on soft membranes.

1.6. Motivation, aim and approach

1.6.1. Motivation and aim

Previous studies have demonstrated that septins can directly bind with the plasma membrane [60, 61, 62] and play an essential role in cellular functions such as cell division [42], acting as diffusive barriers [44, 45] and molecular scaffolds [28]. Furthermore, it has been shown that septins are necessary for the mechanical integrity and motility of cells [44, 48], can associate with other cytoskeletal components [4], and respond to or generate membrane deformations [33, 41], revealing a more dynamic role than previously considered. Recent studies have shown that septins can also form stiff and ductile membrane-associated meshes *in vitro* [33, 36]. These results in combination with Laplaud's observations [27], discussed in Section 1.2, indicate that a septin cortical network could be present in eukaryotic cells, directly affecting membrane properties such as membrane rigidity, even in absence of an actin network.

The present project consists of an experimental investigation of human septin-membrane interactions towards the reconstitution of such a septin cortical network inside free-floating membranes emulating the cellular geometry. To this end, we encapsulate human hexameric and octameric complexes in GUVs and aim to:

- Quantify septin localization in relation to the lipid membrane content
- Assess the effect of septin-membrane interactions on membrane mechanics

Investigation of the septin-lipid affinity can help us establish the conditions under which a membrane-associated septin meshwork can form and identify possible differences in the biophysical properties of human hexameric and octameric complexes. Finally, the mechanical characterization of GUVs encapsulating septins can provide insights in the role septin plays in cell cortex functionality.

1.6.2. Approach

Our experimental model system consists of GUVs of different membrane compositions encapsulating human septin hexamers and octamers. We prepare the system using emulsion Droplet Interface Crossing Encapsulation (eDICE) [54], a variation of the double-emulsion based technique of GUV production, cDICE [53, 68]. After production, we image the GUVs with confocal microscopy for the quantification of septin localization on the membrane and mechanically characterize the GUV membranes with Vesicle Fluctuation Analysis (VFA) using epifluorescence microscopy.

System preparation

eDICE is an emulsion-based technique that allows the production of GUVs encapsulating an inner aqueous solution (IAS) that can differ from the outer aqueous solution (OAS) (Fig. 1.6). Firstly, the lipids that will form the GUV membrane are dissolved in an oil mix. Part of the lipid-in-oil solution is kept aside and the rest is added in a spinning chamber that contains the OAS, leading to the formation of a lipid monolayer at the phase interface. Then, the IAS that contains the proteins to be encapsulated and a higher density medium is added in the remaining lipid-in-oil solution in an eppendorf tube that is then rapidly rubbed on a rough surface to form inverse emulsion droplets surrounded by a lipid monolayer. This pre-formed emulsion is then added to the spinning chamber. Subsequently, due to the centrifugal force and the density gradient, the emulsion droplets eventually cross the lipid monolayer, leading to the formation of GUVs as the lipid monolayers zip together. Lipid preparation and eDICE took place under low humidity conditions, as they have been reported to improve sample quality [68].

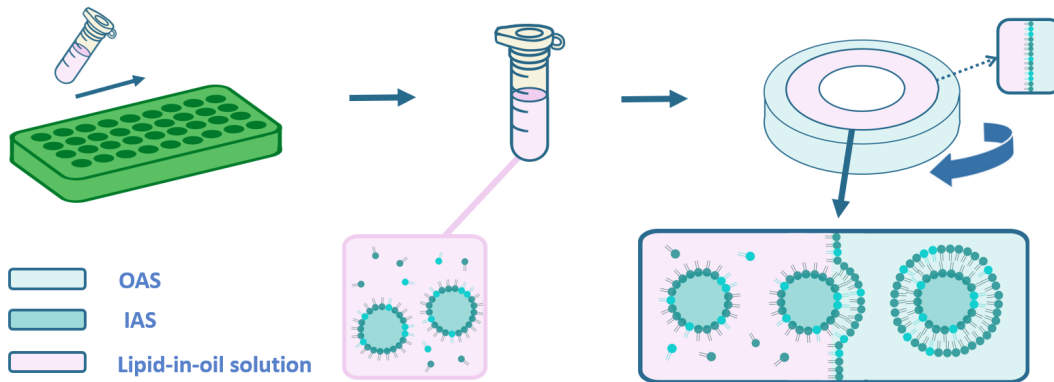


Figure 1.6.: **Schematic of the eDICE process for GUV production.** After addition of the IAS containing the septins in a lipid-in-oil solution, rapid rubbing on a rough surface leads to the formation of inverse emulsion droplets. The pre-formed droplets are then added in a spinning chamber where a lipid monolayer has previously formed between the OAS and the remaining lipid-in-oil solution. The centrifugal forces push the droplets to cross the interface leading to the formation of GUVs that encapsulate the protein of interest.

1. Introduction

The use of pre-formed emulsions in the eDICE protocol notably eases the vesicle production process compared to cDICE, as it removes the need for a capillary that can potentially be clogged and significantly limits the time interval between the IAS addition and GUV formation, avoiding premature polymerization of filamentous proteins such as septins. Finally, it requires smaller IAS volumes ($< 30 \mu\text{L}$ compared to $>100 \mu\text{L}$), easing work with recombinant proteins [54].

Mechanical measurements

There are two main categories of membrane mechanical measurements, either by avoiding the application of any external stresses and studying the thermally excited membrane fluctuations [69, 70] or through an externally modulated membrane deformation induced through micropipettes, optical tweezers, or electric/magnetic fields [51]. These methods are often developed particularly for free-floating GUVs, where no substrates or neighboring membranes interfere with the lipid bilayer [20, 69, 71]. In our work, we focus on Vesicle Fluctuation Analysis (VFA) using epifluorescence microscopy to mechanically characterize the GUV membranes.

VFA is a non-invasive experimental technique that allows the calculation of the membrane's bending rigidity, κ , and tension, σ . As lipid bilayers are assembled by non-covalent interactions and have a thickness of a few nanometers, they are significantly deformable and susceptible to spontaneous thermally induced elastic bending fluctuations, also called undulations [72]. When the bilayer is under low tension, the relevant length scales associated with the fluctuations can be observed with light microscopy [73] and, as they are determined by the elastic moduli of the bilayer, they allow for the experimental determination of the bending modulus, κ , and the membrane tension, σ [20, 69, 70].

2. Materials and Methods

2.1. Septin preparation

Experiments were conducted with human septin hexamers of the structure: SEPT2-SEPT6-SEPT7-SEPT7-SEPT6-SEPT2 and human septin octamers of the structure: SEPT2-SEPT6-SEPT7-SEPT9-SEPT9-SEPT7-SEPT6-SEPT2, with the SEPT9 isoform SEPT9i1. To enable imaging, 15% of the oligomers used in each sample were fluorescently labelled with a monomeric superfolder green fluorescent protein (msfGFP) on the C-terminus of SEPT2.

Both GFP-labeled and unlabeled septins were purified by our lab technician, Jeffrey den Haan, using a double affinity chromatography approach [37, 74]. This method allows for the isolation of septin complexes containing both His6-tagged SEPT2 and Strep-II-tagged SEPT7 or SEPT9 to obtain hexamers or octamers, respectively. Plasmids were co-transformed in *E.Coli* for protein production. After growth and protein expression, the cells were lysed and the cell lysate was clarified by centrifuging. The desired septin complexes were then isolated with two consecutive steps of affinity chromatography, first in a Nickel column and then in a Strep-tag-column. Subsequently, the purified septin complexes were dialyzed in 20mM Tris-HCl at 7.4 pH, 2mM MgCl₂ and 300mM KCl, to avoid septin polymerization, and concentrated. The proteins were then snap-frozen and stored at -80°C. On the day of the experiment, they were thawed and kept on ice until encapsulation in GUVs with eDICE.

2.2. Lipids

All lipids used were acquired from Avanti Polar Lipids. The basic component for the vesicle membranes used was either 1,2-dioleoyl-sn-glycero-3-phosphocholine (DOPC) or 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine (POPC). DOPC and POPC were used interchangeably as they only differ in the saturation of the alkyl chains that are within the hydrophobic core of the GUV bilayer and are not expected to interact with the encapsulated proteins, while the same headgroup is exposed to the aqueous solution.

To investigate septin-lipid affinity, membranes of different compositions were prepared containing the negative lipids required for septin-membrane interactions [33, 36] 1,2-Dioleoyl-sn-glycero-3-phospho-L-serine (DOPS) and L- α -phosphatidylinositol-4,5-bisphosphate (Brain, Porcine) (ammonium salt) (PIP2) in different ratios (Table 2.1). All different compositions tested contained 0.2% 1-oleoyl-2-(12-biotinyl (amino-dodecanoyl))-sn-glycero-3-phosphoethanolamine (Biotin-PE), to assist vesicle binding on the treated imaging surface, and 0.05% 1,2-dioleoyl-sn-glycero-3-phosphoethanolamine-N-(lissamine rhodamine B sulfonyl) (Rhodamine-PE) as a fluorescent label.

All lipids were stored at -20°C in chloroform under Argon, except PIP2 which was stored in 20:9:1 v/v chloroform:methanol:water under Argon for improved solubility.

2.3. GUV preparation

2.3.1. Buffers

To prepare an IAS containing 800nM of septins in conditions that promote septin polymerization (50mM KCl), but avoid polymerization prior to septin encapsulation in the GUVs, we first prepared a Septin mix in six times the desired concentration and a 5×Septin Polymerization Buffer (5×SPB). The Septin mix and the 5×SPB were then mixed to obtain the final IAS.

The Septin mix contained 15% GFP-labelled and 85% unlabelled septin oligomers at a concentration of 4.8μM, in Septin Buffer (20mM Tris-HCl at 7.4 pH, 2mM MgCl₂, 300mM KCl with 1mM DTT) (Tables 2.2 and 2.3). The 5×SPB contained 83.33mM Tris-HCl at 7.4 pH, 8.33mM MgCl₂, 5mM DTT, 5mM GTP, 2.5mM MgATP, 5mM Trolox and 5mM PCA (Table 2.4). 27uL of IAS were prepared per sample. The IAS contained 1×SPB (5.4uL of 5×SPB), 6.5% OptiPrep (1.75uL), 0.05mM PCD (0.27uL), Milli-Q water (6.075uL), G-Buffer (5mM Tris-HCl 7.8 pH, 0.2mM CaCl₂, 9uL) and 800nM Septin mix (4.5uL). The Septin mix was added immediately prior to the emulsion droplet formation to avoid pre-mature polymerization.

Tris-HCl at 7.4 pH was included to control the pH of the solution, MgCl₂ to assist GTP-septin binding and DTT to keep cysteine reduced. The amounts of Tris-HCl and MgCl₂ included in the 5×SPB were added to maintain concentrations of 20mM and 2mM, respectively, in the final completed IAS. GTP was added to promote septin filament assembly and OptiPrep as a density gradient medium for the emulsion droplets to cross the interface with centrifugation. MgATP and G-Buffer were also included to allow for consistent comparison with experiments that also include actin encapsulation. The photoprotective agent Trolox [75] was incorporated to protect from fluorescence blinking and PCA to provide a substrate for PCD which acts as an oxygen scavenger for improved dye stability [76].

The osmolarity of the IAS was measured using a freezing point osmometer (Osmomat 3000, Gonotec) and found to be 176 ± 5 mOsm/kg. All subsequent osmolarity measurements were conducted with the same device and the reported corresponding errors are the standard deviations of three separate measurements. An aqueous solution of 190mM glucose was prepared for the OAS. Its osmolarity was measured to be 188 ± 2 mOsm/kg. After GUV formation, a second aqueous solution, referred to as the Vesicle buffer (VB), and composed of 20mM glucose and 80mM Tris-HCl of 7.4 pH, was added to the OAS to control the pH of the medium. The Vesicle buffer was added after the vesicle formation as it has been observed to negatively affect it, possibly due to disturbances during the formation of the lipid monolayer at the oil-water interface. As the Vesicle Buffer osmolarity was 199 ± 3 mOsm/kg and the solutions were mixed in a 3:1 ratio OAS:VB, the estimated osmolarity of the vesicles medium is 191 mOsm/kg, approximately 15 mOsm/kg higher than the osmolarity of the IAS, ensuring that the vesicles are slightly deflated.

2.3. GUV preparation

Component	Membrane composition (molar ratios %)					
	A	B	C	D	E	F
DOPC/POPC	99.75	94.75	79.75	78.75	76.75	74.75
DOPS	0	0	20	20	20	20
PIP2	0	5	0	1	3	5
Biotin-PE	0.2	0.2	0.2	0.2	0.2	0.2
Rhodamine-PE	0.05	0.05	0.05	0.05	0.05	0.05

Table 2.1.: The six different lipid compositions of GUV membranes investigated in this study.

Septin Buffer		
Component	Concentration (mM)	Function
Tris-HCl 7.4 pH	20	pH control
MgCl ₂	2	Assists GTP-septin binding
KCl	300	Inhibits septin polymerization
DTT	1	Keeps cysteine reduced

Table 2.2.: The composition of the Septin Buffer and the functions of the individual components.

Septin mix	
Component	Concentration (nM)
Dark septins (85%)	6 × 680
msfGFP septins (15%)	6 × 120
Septin Buffer	
Total septin concentration:	6 × 800 nM

Table 2.3.: The composition of the septin mix. The mix was prepared with 6 times the desired final septin concentration and kept in presence of 300mM KCl to prevent premature polymerization until 1:6 dilution in the IAS.

2. Materials and Methods

5×Septin Polymerization Buffer		
Component	Concentration (mM)	Function
Tris-HCl 7.4 pH	83.33	pH control
MgCl ₂	8.33	Assists GTP-septin binding
DTT	5×1	Keeps cysteine reduced
GTP	5×1	Promotes filament assembly
Trolox	5×1	Protects from blinking effect
PCA	5×1	Substrate for PCD
MgATP	5×0.5	Consistency with actin encapsulation

Table 2.4.: The composition of the 5×Septin Polymerization Buffer and the functions of its different components. The 5×SPB was then diluted 5 times in the IAS. The amounts of Tris-HCl 7.4 pH and MgCl₂ added ensure that the final corresponding concentrations in the IAS will remain 20mM and 2mM respectively.

IAS			
Component	Concentration (mM)	Volume (uL)	Function
5× SPB	1×	5.4	
OptiPrep	6.5%	1.755	Density gradient medium
PCD	0.05	0.27	Oxygen scavenger
Septin mix	0.8	4.5	Encapsulated protein
Milli-Q		6.075	
G-Buffer		9	Consistency with actin encapsulation
Total prepared volume per sample:		27	

Table 2.5.: The composition of the completed IAS and the function of its individual components.

2.3.2. Lipid-in-oil solution

The day before an experiment, the different lipids were mixed in glass vials in the desired ratios for a final total concentration of 0.2 mg/ml in a final volume of 7ml. Then they were blow-dried with a nitrogen gun and left under vacuum in a desiccator overnight to remove the solvents and avoid any moisture in the mixture that would undermine the quality of the formed GUVs [68]. For the same reason teflon tape was placed around the screw of the glass vials for tighter sealing once the cap is on. To avoid dye photo-bleaching, before placement in the desiccator, the vials were covered in aluminum foil. After drying, the lipid vials were kept covered at room temperature and used within the same day.

The vials with the dried lipid films were subsequently placed in a glovebox of <1% environmental humidity for re-suspension, as it has previously been demonstrated that high humidity levels during this stage can have a negative effect on the quality of the produced GUVs [68]. An oil mix of silicone oil and mineral oil in a volumetric ratio of 5.3:1.2 was prepared and thoroughly mixed. The dried lipid films were then dissolved in 50uL chloroform and then diluted with 400uL of decane for improved vesicle quality [68]. Finally, 6.5mL of the oil mix were added to each lipid vial under continuous vortexing. Upon removal from the glovebox, the lipid vials were sealed with parafilm and sonicated for 15 minutes in an ice bath using a bath sonicator (Branson 2510 Ultrasonic Cleaner, Marshall Scientific).

2.3.3. eDICE

Home-built spinning chambers as described in [68] were rotated at 2000rpm. Initially, 700uL of OAS were added, followed by 5ml of the lipids-in-oil solution. 1ml of the lipids-in-oil solution was also pipetted into an 2ml eppendorf tube for the preparation of the emulsion droplets. At that stage, the Septin mix was finally added to the IAS, and immediately 25uL of the IAS were pipetted in the 2mL eppendorf tube, which was then rapidly rubbed over an eppendorf holder 12-15 times to create the emulsion through the strong induced vibrations. 1mL of the emulsion was then quickly pipetted into the spinning disc and centrifuged for 3 minutes, so that the droplets have enough time to cross the lipid monolayer at the phase interface for the formation of the GUVs.

After turning off the voltage to stop the chamber from spinning, most of the oil phase was carefully removed from the edge of the chamber's opening with simultaneous gradual tilting of the disc. Then, 233uL of Vesicle buffer were added at the aqueous phase for a final concentration of Tris-HCl of 20mM to stabilize the pH. The discs were then left at a 45° angle for 15 to 30 minutes for the GUVs to sink. For picking up the GUVs, a 200uL pipette tip was cut to minimize shear forces and an air bubble was released upon passing through the interface to remove oil from the tip. After picking up the GUVs, the pipette tip was wiped with paper to remove any oil and then the GUVs were pipetted in an eppendorf tube and left there for approximately 5 minutes, before they were put in imaging wells, as specified in the following sections.

2.4. Quantification of septin encapsulation efficiency

To quantify the septin encapsulation efficiency of GUVs prepared with eDICE, GUVs of DOPC membranes encapsulating septin hexamers were prepared within the same day with septin hexamer bulk solutions of different concentrations and imaged with confocal microscopy. The samples were placed in different wells of the same imaging slides and were imaged under the same conditions to avoid any possible artifacts during the imaging process.

2.4.1. Sample preparation

GUVs composed of DOPC:Biotin-PE:Rhodamine-PE at a molar ratio of 99.75:0.2:0.05 and encapsulating 800nM septin hexamers were prepared with eDICE as described in Section 2.3, using a modified IAS (20mM Tris in pH 7.4, 2mM MgCl₂, 300mM KCl, 1mM DTT, 1mM Trolox, 1mM PCA, 0.05mM PCD and 6.5% OptiPrep) to maintain high salt concentration and inhibit septin polymerization inside the GUVs. The modified IAS osmolarity was measured to be 604 ± 3 mOsm/kg and therefore an OAS of 625mM glucose and a Vesicle buffer of 475mM glucose and 80mM Tris-pH7.4 with 628 ± 3 mOsm/kg and 651 ± 2 mOsm/kg osmolarity, respectively, were prepared, for an approximate osmolarity difference of 30 mOsm/kg between the inner and the outer aqueous solution.

To provide a calibration curve, septin hexamer bulk solutions of six different septin concentrations (0 , 200, 400, 600, 800, 1000 nM) were prepared with 15% of the septin fluorescently labelled with msfGFP in a buffer of the same composition as the new modified IAS. The septin bulk solutions were initially prepared with double concentrations and were diluted twofold within the imaging wells. Prior to the preparation of the solutions, septin stocks were airfuged for 15 minutes under 149000 g (Air-driven Ultracentrifuged, Beckman Coulter) to prevent pre-formed septin aggregates from appearing in the septin bulk solutions.

To prevent septin from binding on the imaging surface the samples were placed on custom-made wells on glass slides (#1 Cover Glass, Corning) covered with neutral Supported Lipid Bilayers (SLBs) formed after Small Unilamellar Vesicle (SUV) fusion with the surface. Initially, the glass slides were cleaned at 70°C for 10 minutes in a base piranha solution (5% hydrogen peroxide, 5% ammonium hydroxide). The slides were then washed with Milli-Q and dried with a nitrogen gun. The wells were prepared by cutting the bottom of PCR tubes and gluing them top-down on the glass slides. The wells were covered with lids of 500uL eppendorf tubes and kept in a humidity box at all times prior to imaging to prevent evaporation.

For the preparation of the SUVs, 99.975:0.025 DOPC:Rhodamine-PE were mixed in a glass vial for a final total concentration of 0.25mM in a final volume of 1ml. Subsequently, the lipid mix was blow-dried with a nitrogen gun. To rehydrate the lipid film and create a suspension of multilamellar vesicles, 1ml of F-Buffer (20mM Tris in pH 7.4, 2mM MgCl₂, 50mM KCl) was added in the glass vial and the solution was vortexed for 1 minute and left for 5 minutes to rest. This process was repeated 4 times. To obtain the SUVs, the suspension was then moved to an eppendorf tube and sonicated using a Bandelin Sonoplus HD2070.2 equipped with a BR30 resonator cup for 30 min in pulsed mode (5 s on/5 s off at 10% amplitude) to avoid heating. The vesicles were stored at room temperature and used within 3 days.

On the day of the experiment, MgCl_2 was added to the SUV solution to get a final concentration of 5mM and then 40uL of the solution were pipetted in each well and left for at least 30 minutes for SUV rupture and SLB formation on the bottom of the well. Subsequently, the wells were washed twice with 100uL of Septin buffer and twice more with 3:1 OAS:VB in the case of the well meant for the GUV sample. The buffers were then partly removed, leaving 20uL to prevent the SLBs from drying. 20uL of the septin bulk solutions were added for 1:2 dilution in the well. Similarly, 20uL of the GUV sample were pipetted in the well containing the 3:1 OAS:VB.

2.4.2. Data acquisition

The samples were imaged on an inverted Leica Stellaris 8 point scanning confocal microscope using a 63x glycerol immersion objective, a white light laser and an HyD S2 detector operated in counting mode. msfGFP was excited using a wavelength of 473nm, 40% laser intensity and line accumulation set to 2. Additionally, the membrane channel was imaged separately (wavelength 573nm) for inspection of the GUVs and the SLBs formed on the bottom of the imaging wells.

To obtain calibration data, 5 images of 2048x2048 px were collected for each one of the septin bulk solutions. The images were all taken at approximately 7 μm height from the imaging surface, determined using the reflection observed in an xzy field of view. Then, at the same height from the surface and using the same imaging settings multiple regions of the GUV sample were collected.

2.4.3. Image Analysis

For each septin bulk solution, the average signal of the 5 collected regions was calculated and, assuming a linear relation between pixel intensity values and septin content of the form: $I_{\text{sept}}(c_{\text{sept}}) = A \cdot c_{\text{sept}}$, the proportionality constant A was determined in Python using the numpy least square fitting to obtain the calibration curve.

The positions and radii of the GUVs were detected using DisGUVery [77], an open-source software specialized in the analysis of GUVs. The collected raw images were imported in the DisGUVery software and pre-processed with a 2D Gaussian smoothing of kernel size 15 and edge enhancement of a kernel size of 30 to facilitate the vesicle detection. Subsequently, the Circular Hough Transform detection method [78] was implemented for the detection of the vesicles (Fig. 2.1A). The parameters selected for the detection were: Edge threshold: 30, Hough threshold: 30, Min. vesicle distance: adjusted per image based on the vesicle density, Min. radius: 10, Max radius: 200 (Fig. 2.1B).

Then, the detected vesicle positions were imported in a custom Python script, where circular ROIs were created, concentric to each GUV and with radii of 90% of the respective vesicle radius (Fig. 2.1C), to avoid underestimation of the septin signal due to misaligned GUV detection. After visual inspection of the drawn ROIs, false detections and GUVs that included large septin aggregates, usually found on deformed areas of the membrane were rejected, as in these cases the homogeneous septin signal in the lumen does not correspond to all the encapsulated content (Fig. 2.1D). For the remaining vesicles, the pixel intensities within the ROI areas were then averaged for each vesicle providing a distribution of encapsulated septin concentration after conversion using the determined proportionality constant A.

2. Materials and Methods

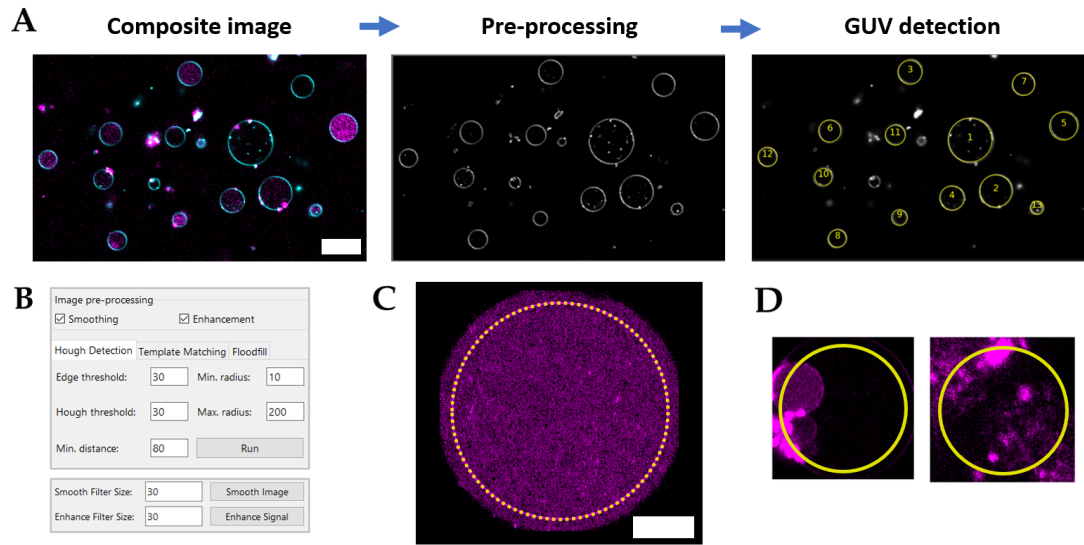


Figure 2.1.: Analysis workflow of GUV samples for the quantification of septin encapsulation. (A) A two-channel confocal image (membrane: cyan, septin: magenta) was acquired at 7 μm height from the imaging surface. The image was pre-processed with Gaussian smoothing and edge enhancement to ease the detection of the vesicles. The vesicle positions and radii were obtained with DisGUVery. Scale bar: 20 μm . (B) The settings used in the DisGUVery software for the pre-processing and the detection with the Circular Hough Transform method. (C) The average intensity of the septin signal was calculated within a circular ROI of 90% the vesicle radius (yellow dashed line) and later converted to concentration using the calibration curve obtained from the septin bulk solutions. Scale bar: 5 μm . (D) Examples of rejected detections due to misalignment or the presence of septin aggregation.

2.5. Quantification of septin localization

2.5.1. Sample preparation

To quantify septin localization on the vesicle membrane and identify septin lipid affinity, septin hexamers and octamers were encapsulated in GUVs of six different membrane compositions (see Table 2.1). The GUV samples were prepared with eDICE as described in Section 2.3 and imaged with confocal microscopy.

2.5.2. Data Acquisition

The samples were placed in ibidi 18-well μ -slides. For GUV immobilization on the imaging surface, the wells were previously passivated by incubation with 1mg/ml of a BSA:BSA-biotin solution in Milli-Q for at least 15 minutes at room temperature, followed by incubation with 1mg/ml of Neutravidin in 3:1 OAS:VB for at least 5 minutes. Subsequently, the wells were washed with 3:1 OAS:VB twice. 40uL of the prepared GUV samples were pipetted in each well containing 40uL of 3:1 OAS:VB for 1:2 dilution.

The GUV samples were imaged on an inverted Leica Stellaris 8 point scanning confocal microscope using a 63x glycerol immersion objective, a white light laser, and HyD detectors operated in analog mode. Rhodamine and msfGFP were excited sequentially at 573 nm and 489 nm, respectively. As the msfGFP signal corresponding to the septin was significantly lower than that of the Rhodamine-labelled lipids, much higher laser intensity and gain percentages were required for imaging and line accumulation was set to 2. The microscope settings are collected in Table 2.6.

For the assessment and quantification of septin localization on the membrane, a series of two-dimensional image regions were collected for each prepared sample. The regions were all taken at approximately 7 μ m height from the imaging surface, determined using the reflection observed in an xzy field of view. This height corresponds approximately to the average vesicle radius, so that more vesicles are at their equatorial plane. Additionally, z-stacks were also collected for a number of vesicles to identify whether localization appears to be uniform for the whole vesicle inner surface.

2.5.3. Image Analysis

A custom-written Python script was developed (available at the repository <https://github.com/kkourkoulou/membrane-localization-quantification>), aiming to quantify the localization of encapsulated septin on the GUV membranes, allowing for comparisons between different membrane compositions. The script uses the position and the size of the vesicles present in each image as determined in DisGUVery [77]. The radial and angular intensity profiles of each vesicle are calculated after background signal subtraction. Following an automatic rejection of vesicles of bad quality or objects mistakenly detected as vesicles with conditions based on the calculated intensity profiles, the script proceeds to the calculation of the localization of septin on the lipid membrane.

2. Materials and Methods

Signal	Wavelength (nm)	Laser intensity (%)	Gain (%)
Rhodamine	573	1.5	2
GFP	489	23	80

Table 2.6.: The microscope settings used for data acquisition for the quantification of septin localization with confocal microscopy.

Vesicle detection

Initially, the raw confocal images corresponding to the membrane channel were imported in the DisGUVery software [77] for vesicle detection. After a pre-processing step of image smoothing by convolution of the original image with a 2D Gaussian function of kernel size 3, the Circular Hough Transform detection method [78] was implemented for the detection of the vesicles. The parameters selected for the vast majority of the images were: Edge threshold: 35, Hough threshold: 35, Min. vesicle distance: 30, Min. radius: 10, Max radius: 80, with slight modifications for improved detection based on the density and clarity of each sample.

Radial intensity profiles and Background signal subtraction

The raw confocal images corresponding to the membrane and the septin channel were then imported in the Python script together with the information of the vesicle positions and radii calculated in DisGUVery. Subsequently, for each vesicle, 360 equally-spaced radial paths were created starting from the detected vesicle centre. The path length was 1.5 times the length of the detected vesicle radius to account for possible membrane deformations and detection misjudgments in the placement or size of the vesicles by DisGUVery. Along each radial path, the signal intensity of both the membrane and the septin channel were measured with a step of a single pixel. The individual radial intensity profiles calculated in this manner were then averaged for all angles to obtain a collective radial profile for the vesicle while at the same time reducing the effect of noise (Fig. 2.2A).

Additionally, background signal was also subtracted from the collective radial profile to obtain the corrected mean radial profile of the vesicle. For the calculation of the background signal, a mask was created locally using the Li threshold in a square of side equal to 3 times the vesicle diameter centered on the vesicle under study. The threshold was applied locally to account for variations in the background signal over the wide region of the considered image. Closed contours of the mask were then filled in to efficiently avoid the signal of the vesicle lumen and of possible surrounding lipid aggregates or segments of other vesicles found in this area. The background signal for each of the channels was then calculated as the average of the corresponding signal intensity in the remaining area (Fig. 2.2B). The Li threshold was chosen after visual inspection of the performance of seven different options available in the filters module of the scikit-image Python package (Isodata, Li, Mean, Minimum, Otsu, Triangle and Yen) for a subset of randomly chosen sample images.

Membrane detection and Angular intensity profiles

From the calculated radial intensity profile, the position of the membrane borders was re-defined using the peak detection function `find_peaks` of the `scipy` Python package, requiring a minimum height of 10 units, a minimum distance from other peaks of 5 units, and a prominence of 1. These parameters were chosen after trials and visual inspection of randomly chosen sample images. The membrane's position was considered to be the position of the detected peak. However, in dense samples the membranes of neighbouring vesicles or lipid aggregates could possibly result in additional prominent peaks visible in the extension of the radial profiles, and thus in the detection of multiple peaks in that region. To account for that possibility and avoid overestimation of the size of the vesicle, the heights of the last two or three (if present) peaks detected outside the vesicle radius found in DisGUVery were compared and the tallest peak was considered to correspond to the membrane position (see Appendix Fig.A.1A). The width of the membrane area was then taken as the chosen peak's width at half-maximum and the inner and outer borders of the membrane were thus determined (Fig. 2.2A). Subsequently, the angular profile of the vesicle along the membrane contour was calculated by averaging the pixel intensities within the detected membrane ring for each angle.

Automatic vesicle rejection and corrections

In order to study the localization of septin on the membrane, it was necessary that the detected and considered vesicles were properly in focus and did not include lipid pockets, significant deformations, or other artifacts that could interact and interfere with the protein e.g. encapsulated lipid aggregates, clusters of small vesicles on the membrane, multilamellar structures or lipid tubes. However, the quality of the prepared samples was found to vary significantly and such vesicles were often encountered, as well as other objects present in the outer buffer such as lipid aggregates, oil droplets, or crowded clusters of vesicles. Consequently, as the Circular Hough Transform method is based on edge and circularity detection, such unwanted objects were also falsely detected.

To ease the process of eliminating these undesired detected structures in the large collected datasets, an automatic rejection process was incorporated in the script. With it, individual vesicles were marked with appropriate comments based on the related type of artifacts reflected in the behaviour of the radial and/or the angular profile (see Appendix Fig.A.1B-H and Fig.A.2). The marked objects were subsequently removed and the remaining GUVs were isolated and analyzed. Detected objects were rejected if any of the following criteria was satisfied:

- Vesicles were crossing the image borders or some of the radial paths for the calculation of the intensity profiles extended further than the image margins (Fig. 2.2C.a).
- No septin signal was present inside the vesicle or septin signal was too low and dominated by noise. Condition chosen after visual inspection: Maximum of the septin signal inside the vesicle was lower than 5 a.u.
- No membrane peaks were detected with the chosen parameters. Intended to reject lipid aggregates, oil droplets and dense lipid clusters that present a rather uniform signal rather than a peak corresponding to clear circular membrane border as in the case of GUVs (Fig. 2.2C.b).

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- The detected membrane was wider than at least 1/4 of the vesicle radius. Intended to reject the vesicles that are out of focus (Fig. 2.2C.c).
- Peaks were detected inside the GUV lumen. Intended to disregard vesicles with encapsulated lipid aggregates that might interact with the encapsulated septin and interfere with septin localization (Fig. 2.2C.d).
- The average lipid signal from the centre to the inner membrane border was larger than 1/3 of the signal at the membrane peak. Intended to avoid vesicles that contain large lipid aggregates or that encapsulate smaller vesicles (Fig. 2.2C.e).
- The average lipid signal outside the outer membrane border was larger than 40% of the signal at the membrane peak. Intended to reject vesicles that are part of large vesicle/lipid clusters (Fig. 2.2C.f-g).
- The relative standard deviation (RSD) of the septin angular intensity profile was higher than 0.8. Intended to avoid vesicles with significant deformations, lipid pockets or large aggregates on the membrane, where it is unclear whether septin is localized due to binding or trapped in lipid compartments (Fig. 2.2C.h and Fig. A.2 in the Appendix). The threshold value of the RSD was chosen through visual inspection, aiming to be able to detect such scenarios but avoid simultaneous rejection of vesicles with no clusters but with high variations in the angular profile due to noise.

Septin localization quantification

After the rejection of unwanted objects and prior to the quantification of localization, an additional correction was necessary to avoid any effects of noise magnification due to multiple counting of the central pixels. From manual observation of the data, the removal of the two central pixels in the radial profile seemed sufficient (Fig. 2.2D).

To assess whether septin localizes at the membrane and quantify the localization, we formulated a definition that considers the normalized difference between the septin signal at the membrane and the septin signal in the lumen:

$$L = \frac{\tilde{I}_m - \bar{I}_c}{\tilde{I}_m} \quad (2.1)$$

where $\tilde{I}_m$ corresponds to the median of the septin intensity signal within the width of the detected membrane and $\bar{I}_c$ corresponds to the average septin signal of a central area of radius equal to 1/4 of the detected vesicle radius (Fig. 2.2E).

We calculate the median of the fluorescent intensity within the detected membrane width, instead of the average value, to avoid overestimation of the amount of localized septin due to potential narrow peaks for specific angles of the angular profile that originate from smaller undetected lipid aggregates on the vesicle membrane. In such regions, it is unknown whether septin localized due to lipid affinity or physical entrapment. Thus, calculation of the median provides us with a value that is representative of the homogeneous septin signal observed for the majority of the membrane segments (see Appendix FigA.2). In contrast, as the septin signal away from the membrane is uniform, taking the average value within a concentric central segment is sufficient.

To accommodate for comparisons between different vesicles of the same or different samples, the calculated difference needed to be normalized. As overall the septin signal was low and was found to vary significantly among GUVs, division by $\bar{I}_c$ would lead to a very wide distribution of values that could also go to infinity, forcing the rejection of the corresponding vesicles. Therefore, the value chosen for the normalization is the median signal at the membrane so that L has an upper bound, for $\bar{I}_c = 0$.

This formulation additionally facilitates a convenient qualitative classification of the vesicles, based on whether septin localization is present or absent. Specifically, when $\tilde{I}_m \leq \bar{I}_c$ or equivalently $L \leq 0$, it is obvious that there is no septin localization on the membrane, while for $0 < L \leq 1$ some septin localization is detected. In order to avoid misjudgement and to only include vesicles with clear homogeneous localization, the threshold value $L = 0.16$ was chosen. This value corresponds to $\tilde{I}_m = 1.2\bar{I}_c$ and was chosen after visual inspection of randomly chosen vesicles in order to avoid false assessments due to the noise present in the lumen area of the radial intensity profiles (Fig. 2.2F). Apart from the classification, the calculated distributions of L values for each composition were statistically compared using the Mann-Whitney U test.

2.6. Vesicle fluctuation analysis

2.6.1. Sample preparation

Aiming to identify the effect of a septin meshwork on the membrane's rigidity, GUVs were prepared with eDICE as described in Section 2.3, encapsulating either septin hexamers or octamers. The prepared vesicle membranes contained DOPC:DOPS:PIP2:Biotin-PE:Rhodamine-PE at a molar ratio 74.7:20.5:0.2:0.1. Even though we imaged free floating vesicles and we did not treat the imaging surface to allow for vesicle binding, Biotin-PE was included for consistency with previous experiments. The amount of Rhodamine-PE was doubled compared to previous samples for improved contrast, easing membrane detection and contour tracking in the analysis. Alternatively, the fluorescently tagged lipid ATTO 655 DOPE was used instead of Rhodamine-PE in an attempt to prevent crosstalk between the membrane and the septin signal.

As empty vesicles of the same composition consistently exhibited a high amount of encapsulated small lipid aggregates, viewed as bright diffusing specks, correct membrane tracking and analysis were not possible. Therefore as a control system, vesicles with membranes of 99.7% DOPC, 0.2% Biotin-PE, 0.1% Rhodamine-PE, encapsulating septin octamers were used. As septin cannot bind to the neutral membrane of the control system, the membrane properties cannot be affected.

An OAS of 190 mM glucose and osmolarity of $188 \pm \text{mOsm/kg}$ and a Vesicle buffer of 20 mM glucose and 80 mM Tris-HCl of 7.4 pH of $199 \pm 3 \text{ mOsm/kg}$ were used for a final osmolarity difference of 15 mOsm/kg inside and outside the vesicles. After picking up 100 μL of the vesicle samples as described in Section 2.3.3, the vesicles were left in an eppendorf tube for approximately 15 minutes, before they were put in the imaging wells, to prevent the formation of lipid tubes, prior to further gradual increase of the osmolarity difference upon solvent evaporation during imaging.

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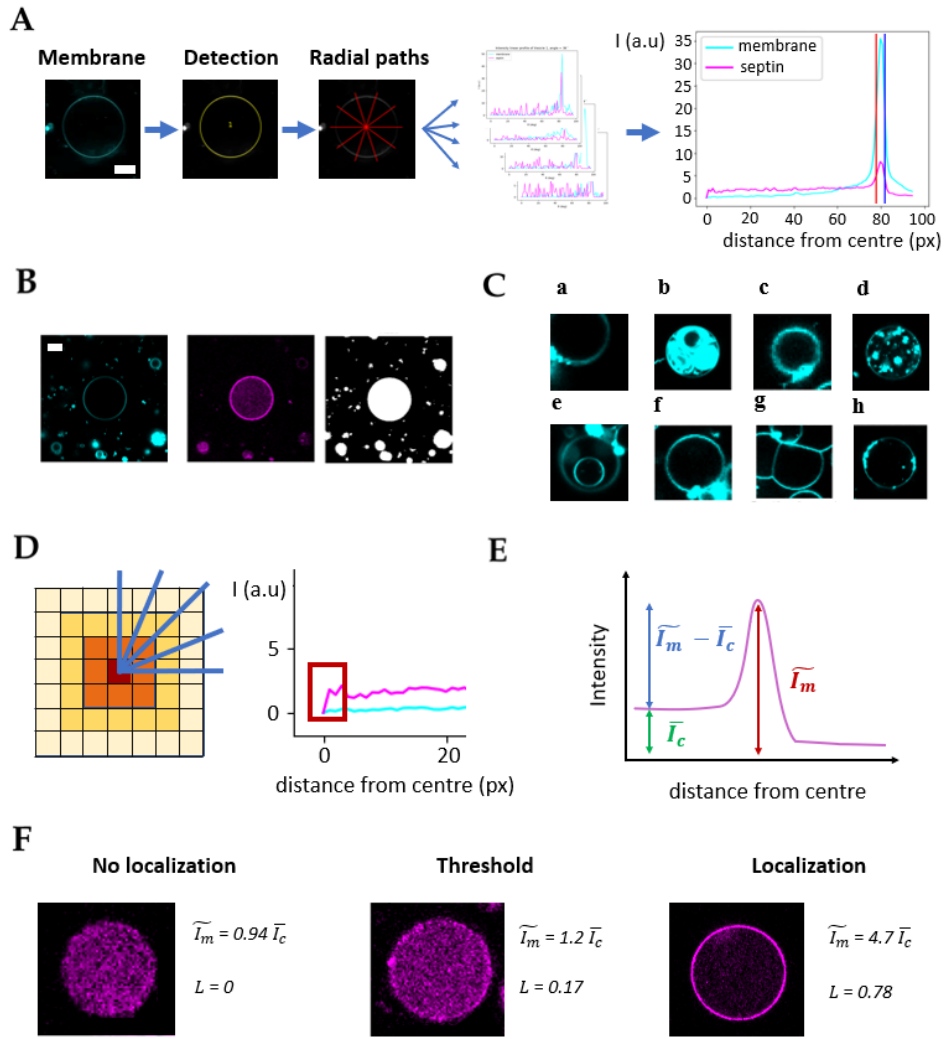


Figure 2.2.: Quantification of septin localization. (A) After detection of the vesicle positions (yellow line), 360 equally spaced radial intensity profiles were calculated (some of them shown in red), and then averaged to obtain a collective radial intensity profile for each vesicle. The borders of the membrane (red and blue lines) were identified as the width at half maximum of the detected peak. Scale bar: 10 μm . (B) A mask (white) was created locally for the calculation of the background noise using the Li threshold. The background of each channel (membrane: cyan, septin: magenta) was calculated by averaging in the unmasked region. Scale bar: 10 μm . (C) Example images of rejected DisGUVery detections. (D) The pixel grid centered at the vesicle center is depicted, with the 1st (red), 2nd (orange), 3rd (yellow), 4th (light yellow) initial pixels. As all radial profiles (blue) originate from the centre, the initial pixels are multi-counted. The effect reduces as the lines grow further from the center and as the number of pixels averaged for a specific radius increases. In the zoomed-in image of the radial profile of (A), the bias from the multi-counting is visible (red rectangle). (E) Visualization of Eq. 2.1. L was defined as the ratio of the blue over the red height. (F) Three different vesicles displaying no localization, signs of localization and clear localization. The value $L = 0.16$ was selected as the classification threshold.

2.6.2. Data Acquisition

The samples were placed in ibidi 18-well μ -slides, previously passivated with the addition of 40uL of β -casein solution (1 mg/mL β -casein, 10 mM Tris-HCl of pH 7.4), left for incubation for at least 15 minutes and blow-dried with a nitrogen gun. Each well was filled with 40uL of 3:1 OAS:Vesicle buffer mix and 10uL of the GUV samples were added for a 1:5 dilution, so that isolated vesicles could be found more easily, allowing for the collection of time-lapse movies with limited interference of neighboring membranes after all vesicles sink to the bottom due to their high density.

The confocal scanning microscope described in Section 2.5.2 has a limited temporal resolution. For this reason, for the fluctuation analysis the vesicles were imaged with a Nikon Eclipse Ti microscope equipped with a digital camera (Orca-Flash 4.0, Hamamatsu) and monochromatic illumination with a LED light source (Spectra X, Lumencor), using a 100x oil immersion objective (Nikon, N.A. 1.40). In epifluorescence mode, time-lapse movies of 2000 frames were recorded at the equator of vesicles at an excitation wavelength of 555 nm (or 640 nm for the experiments with ATTO 655 DOPE) for the membrane channel and 470 nm for the septin channel with 50% and 75% LED intensities, respectively, with exposure times of 5 ms, 10 ms, 20 ms and 50 ms. To increase the number of fluctuating vesicles, the lid of the imaging chamber was removed for approximately 15-20 minutes to gradually increase the osmolarity of the sample's buffer through evaporation.

2.6.3. Image analysis

After acquisition of the time-lapse movies at the equatorial plane of the fluctuating vesicles, we employed a custom-written Python script made by Lennard van Buren [79] for the calculation of the bending rigidity and the membrane tension of each vesicle. The analysis of the data was based on five main steps: pre-processing to accommodate for membrane detection, tracking of the vesicle contour in time, calculating the corresponding fluctuation spectrum, selecting the appropriate mode regime for fitting to the theoretical spectrum and, finally, implementing the fitting for the acquisition of the membrane properties.

Pre-processing

Initially, each vesicle time-lapse movie was pre-processed in Fiji [80]. The time-stacks were cropped to isolate the vesicle from surrounding structures. Then, the vesicle was centered in the image stacks with the use of the StackReg plugin, to prevent possible spatial translations from affecting the tracking of the vesicle contour through time. Subsequently, the resulting time-stacks were loaded in the Python script. A 2D Gaussian smoothing filter (kernel size: 3, standard deviation: 3) was applied to the image to ease contour tracking.

Contour detection

The vesicle contour was determined for the first frame using 360 equally spaced radial profiles, starting from the center of the image and extending past the vesicle membrane. The membrane position for each of the 360 angles was assumed to be the location of the maximum of each profile within a narrow ring of inner and outer radii predefined by the

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user to reduce computation time (Fig. 2.3A). Afterwards, the 360 obtained position vectors were added to the initial guess for the vesicle's center to refine it and the process was repeated until the new center was no further than one pixel from the previous guess. Using the final obtained position the process was repeated for all frames yielding the evolution of the membrane contour $r(\theta)$ through time (Fig. 2.3C).

Calculation of the experimentally obtained fluctuation spectrum

Subsequently, the membrane displacement normal to the membrane (z-direction) due to the thermal fluctuations, $u(z)$, was quantified with respect to the membrane's average position (Fig. 2.3B). To this end, following the methodology of Pécrciaux et al. [69], a discrete Fourier transformation of the obtained vesicle contour was performed:

$$r(\theta) = R \left(1 + \sum_{q=1}^{\infty} a_q \cos(q\theta) + b_q \sin(q\theta) \right) \quad (2.2)$$

where R is the radius of the contour that can be approximated by the average radius over time for all frames, $\langle R \rangle$. After rescaling each point of the contour by $\langle R \rangle$, the modes of the Fourier-expanded contour can then be determined from the expressions [70]:

$$a_q = \frac{1}{2\pi} \sum_i \{ r_i \cos(q\theta_i) + r_{i+1} \cos(q\theta_{i+1}) \} (\theta_{i+1} - \theta_i) \quad (2.3)$$

$$b_q = \frac{1}{2\pi} \sum_i \{ r_i \sin(q\theta_i) + r_{i+1} \sin(q\theta_{i+1}) \} (\theta_{i+1} - \theta_i) \quad (2.4)$$

where q denotes the mode number and, thus, the corresponding wave number, q_x is equal to $q/\langle R \rangle$. Taking the Fourier transform, $\tilde{u}_{q_x}$, of the membrane displacement, $u(z)$ (Fig. 2.3B), the variance of the fluctuation amplitudes could be determined through the variance of the Fourier amplitudes, $c_q = \sqrt{a_q^2 + b_q^2}$ from the relation [69]:

$$\langle |\tilde{u}(q_{x,y=0})|^2 \rangle = \frac{\pi \langle R \rangle^3}{2} \left(\langle |c_q|^2 \rangle - \langle |c_q| \rangle^2 \right) \quad (2.5)$$

Fitting with the theoretical fluctuation spectrum

Although GUVs are spherical, it has been demonstrated that thermal undulations can be described by the equations corresponding to planar membranes, with errors being significant only for the first modes [69]. The fluctuation spectrum of a planar membrane was calculated by Helfrich [72]:

$$\langle |u(\mathbf{q}_{\perp})|^2 \rangle = \frac{k_B T}{\sigma q_{\perp}^2 + \kappa q_{\perp}^4} \quad (2.6)$$

where $u(\mathbf{r}_\perp)$ is the displacement in the z direction, normal to the membrane with respect to its mean position and $\mathbf{r}_\perp = (x, y)$ are the coordinates of the projection of a point of the membrane on the xy -plane. $\mathbf{q}_\perp = (q_x, q_y)$, with its norm denoted as $q_\perp$, is the wave vector corresponding to $\mathbf{r}_\perp = (x, y)$, σ is the membrane tension, κ the bending modulus, k_B the Boltzmann constant and T the absolute temperature. As the time-evolution of the contours corresponds only to the equatorial plane of the vesicle, the theoretical fluctuation spectrum corresponding to the fluctuations measured is:

$$\langle |u(q_{x,y=0})|^2 \rangle = \frac{k_B T}{2\sigma} \left[\frac{1}{q_x} - \frac{1}{\sqrt{\frac{\sigma}{\kappa} + q_x^2}} \right] \quad (2.7)$$

As experimentally the time evolution videos collected for each vesicle consist of frames, fluctuations shorter than the camera exposure time cannot be identified and even for low modes (q_x) the spectrum can still be affected [79]. Therefore, modifications in the theoretical spectrum had to be made to accommodate for proper fitting of the experimental spectrum. We followed the approach proposed by Pecreaux et al. [69], where the experimentally obtained fluctuation spectrum to account for the finite integration time is fitted to the expression:

$$\langle |\tilde{u}_{q_x}|^2 \rangle = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{k_B T}{4\eta q_\perp} \tau_m \frac{\tau_m^2(q_\perp)}{\tau^2} \times \left[\frac{\tau}{\tau_m(q_\perp)} + \exp\left(-\frac{\tau}{\tau_m(q_\perp)}\right) - 1 \right] dq_y \quad (2.8)$$

where $\eta=0.001$ kg/ms is the medium viscosity and τ_m the fluctuation lifetime of a mode $\mathbf{q}_\perp$ given by:

$$\tau_m(\mathbf{q}_\perp)^{-1} = \frac{1}{4\eta q_\perp} \left(\sigma q_\perp^2 + \kappa q_\perp^4 \right) \quad (2.9)$$

Mode selection

To accurately determine the bending rigidity, only a specific mode regime is appropriate for the fitting of the experimental fluctuation spectrum to the theoretical one [69, 70]. Low q modes are typically dominated by the membrane tension and not the bending rigidity, and correspond to long fluctuation lifetimes that do not have time to equilibrate on the time scale of our obtained time-lapse movies. On the other hand, higher q modes are affected by the spatial resolution of the microscope, the integration time and noise and cannot reliably be used for the derivation of κ [79]. Therefore, only the intermediate modes should be considered in the analysis. The regime of the intermediate modes can be identified by plotting the product of the mean squared fluctuation amplitude with the cube of the wave vector, $\langle |\tilde{u}_q|^2 \rangle q_x^3$, as a function of the wave vector, q_x [70]. As in the intermediate regime the variances of the fluctuation amplitudes scale as the inverse cube of the wave number, the corresponding modes can be determined by the position of the appearing plateau (Fig. 2.3D).

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However, the range of modes for a reliable fitting is further reduced due to limitations of the integration time, as fluctuations with shorter lifetimes cannot be accurately fitted. A higher cut-off mode must, thus, be considered given by the relation [69]:

$$q_{high} = \sqrt[3]{\frac{4\eta}{\kappa\tau}} \quad (2.10)$$

where τ is the exposure time, η the solution's viscosity and κ the bending rigidity. In our analysis, assuming a κ of magnitude 10^{-19} J and a viscosity of 0.001 kg/ms, the higher cut-off mode is initially calculated based on the exposure time and considered as the higher border of the fitting for all further analysis.

After the calculation of the experimental spectrum, the variances of the fluctuation amplitudes (Eq. 2.5) of the vesicle are fitted to the theoretical spectrum (Eq. 2.8) using the `optimize.curve_fit` function of the `scipy` Python library. To eliminate subjectivity in the choice of the lower cut-off mode, multiple fits were carried out for all q_{low} values from 1 to q_{high} . For each of these selections, the respective κ were obtained and plotted against the q_{low} choice. The range of q_{low} choices that minimally affect the obtained value for κ were then determined (q_{low}^l, q_{high}^h). The final value for the bending rigidity was then found by averaging the different κ values in that region and the corresponding error was estimated as the respective standard deviation (Fig. 2.3E). Using the κ value determined in this manner and q_{low}^l as the final choice of the lower cut-off mode, the fitting was repeated to obtain the value of the membrane tension, σ (Fig. 2.3F).

Vesicle rejections

All vesicles imaged had a radius in the range of 4 to 10 μm . To avoid the influence of artifacts in the determination of the mechanical properties, time-lapse movies were collected only for vesicles that were clearly separated from other vesicles and lipid aggregates. Additionally, vesicles were imaged only if there were no visible lipid tubes, other secondary structures or deformations on their membranes. As quite often small lipid aggregates would diffuse around the focal plane, either outside or within the vesicles, occasionally appearing close to the membrane contour, during processing time-lapse movies were cut short accordingly and only videos with 500 or more remaining frames were analyzed further. Furthermore, the evolution of the contours length was monitored as an indication of the membrane detection quality and only vesicles with deviations no larger than 2% from the time averaged contour length were accepted, as larger variances were assumed to originate from misjudged detection of bright objects that appeared close to the membrane. Finally, vesicles for which there was no consistency in the κ values obtained with different choices of the lower cut-off mode were also rejected as any choice would be subjective.

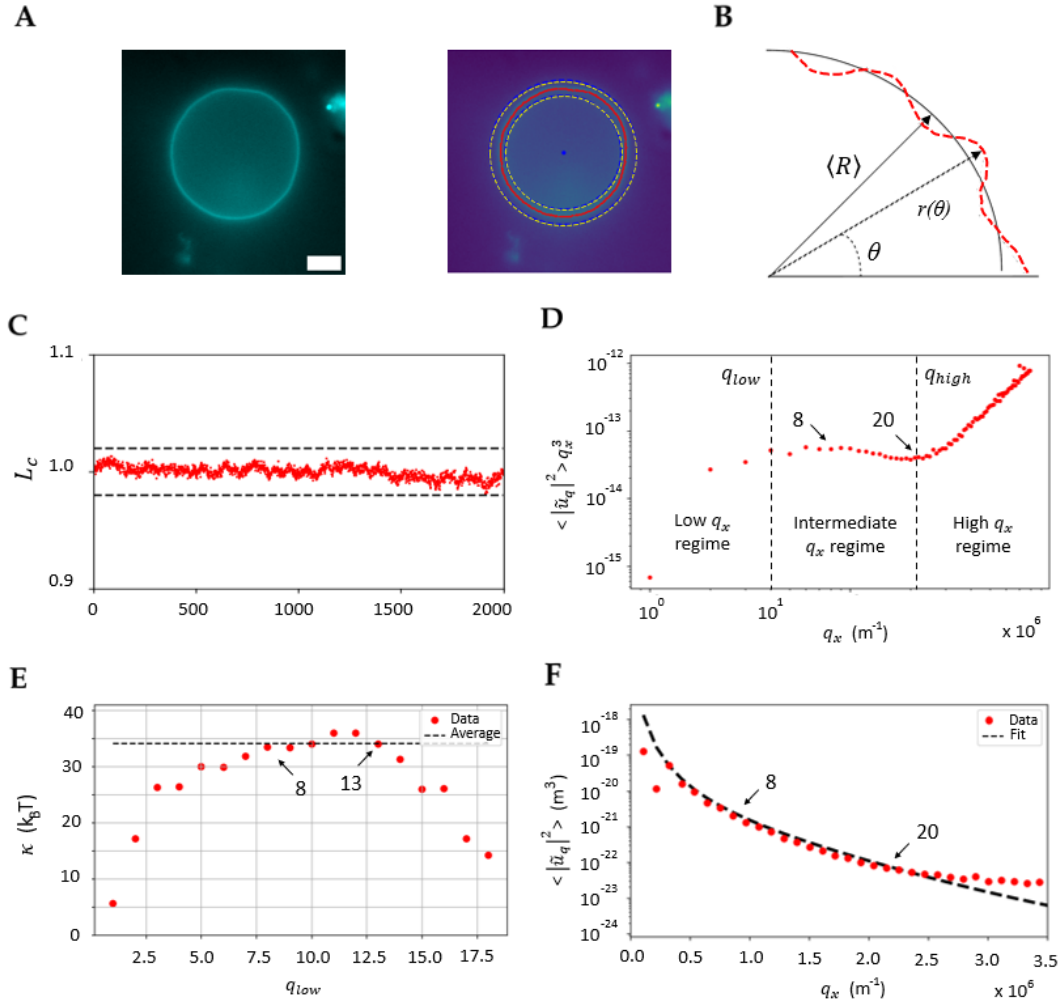


Figure 2.3.: **Workflow of the VFA analysis** (A) Example image of a fluctuating 75:20:5 DOPC:DOPS:PIP2 vesicle. The pre-processed image (left) is loaded in the script for the detection of the contour. The membrane positions are determined for 360 angles from the maximum intensity value (red points) within two rings defined by the user (yellow dashed lines). The image corresponds to the first frame of a time-lapse video. Scale bar: 5 μm . (B) Schematic of a segment of the vesicle membrane, depicting the average vesicle radius over time $\langle R \rangle$ and the vesicle contour in one frame $r(\theta)$ (red dashed line). $u(z)$ corresponds to $r(\theta) - \langle R \rangle$. Figure adapted from [69]. (C) Evolution of the detected contour length throughout the frames. The values are normalized with the contour time average. Dashed lines indicate the $\pm 2\%$ limits for accepted deviations used as a quality control for the membrane detection. (D) Three different mode regimes can be seen by scaling the product of the mean squared fluctuation amplitude with the cube of the wave vector, $\langle |\tilde{u}_q|^2 \rangle q_x^3$, with q_x . Only the intermediate regime is appropriate for fitting. Dashed lines indicate the borders of the plateau corresponding to the intermediate regime and the arrows point at the modes 8 and 20 chosen for the lower and upper cut-off from step (E) and (Eq.2.10), respectively. (E) Different choice of the lower cut-off yields different κ values. In this case, the regime where κ is minimally affected by that choice is between the 8th and the 13th mode. By averaging for these modes κ is determined as $34.1 \pm 1.3 k_B T$, with the error corresponding to the standard deviation. (F) The final fitting using the obtained κ and $q_{low}=8$ yields $\sigma=2.8 \cdot 10^{-9}$ N/m. The lower and higher modes considered are indicated with the arrows.

3. Results

3.1. Evaluation of the eDICE GUV samples

3.1.1. Sample quality

In order to assess the quality of the GUVs prepared with eDICE, we imaged the samples with confocal microscopy and inspected the GUV yield, size and quality of vesicle membranes, as well as the septin encapsulation efficiency. Fig. 3.1A shows a typical field of view of a GUV sample, with the lipid channel shown in cyan and the septin channel in magenta. Even though vesicle density varied per sample (Tables A.1 and A.2), eDICE generally produced a high yield of GUVs, allowing for easy detection of 100-1000 vesicles per imaging well. Despite the observed variability in sample quality and the manual preparation of the emulsion droplets, the produced GUV radius distributions were quite consistent, with an average vesicle radius of approximately 7 μm (Fig. 3.2). While occasionally large septin aggregates could be found in some regions of the collected images (Fig. 3.1B), septin was successfully encapsulated in the GUVs for the majority of the prepared samples (see Section 3.1.2).

Apart from GUVs, other lipid structures and objects would commonly be observed in the imaging chambers. In all produced samples, free-floating lipid aggregates, seen as bright fluorescent specks (Fig. 3.1C), were always visible, while sometimes oil droplets (Fig. 3.1D) were also observed. Often, large immobile lipid aggregates, of sizes ranging from 10 to 100s of μm , appeared attached to the vesicles (Fig. 3.1E). Multilamellar structures were often encountered among the GUVs (Fig. 3.1F), while long lipid tubes would sometimes be seen (Fig. 3.1G).

Although a large number of vesicles was found per sample, few of them were free of other secondary lipid structures present in their membrane and lumen (Fig. 3.1H). Very often, small bright lipid aggregates would diffuse around within the GUVs' lumen (Fig. 3.1I) or smaller vesicles would be encapsulated within the GUVs (Fig. 3.1J). On occasion, the membranes of the GUVs would be attached to neighboring vesicles and lipid aggregates forming clusters (Fig. 3.1E), or would present lipid pockets (Fig. 3.1K) or other secondary membrane structures observed as bright spots on the GUV bilayer (Fig. 3.1L). In the latter case, septin would often appear to localize in these regions independent of membrane composition, possibly trapped within the formed lipid structures.

The percentage of the artifacts described in the previous two paragraphs could vary significantly per sample. Additionally, sometimes samples would yield very few GUVs or GUVs with large assortments of artifacts, even though they were prepared in parallel and under the same conditions with samples that had a high yield of clean-looking GUVs. This is an indication that eDICE is highly sensitive to environmental conditions and manual handling during preparation. To compensate, multiple samples were prepared for each membrane composition of interest. To accommodate for further analysis without the interference of

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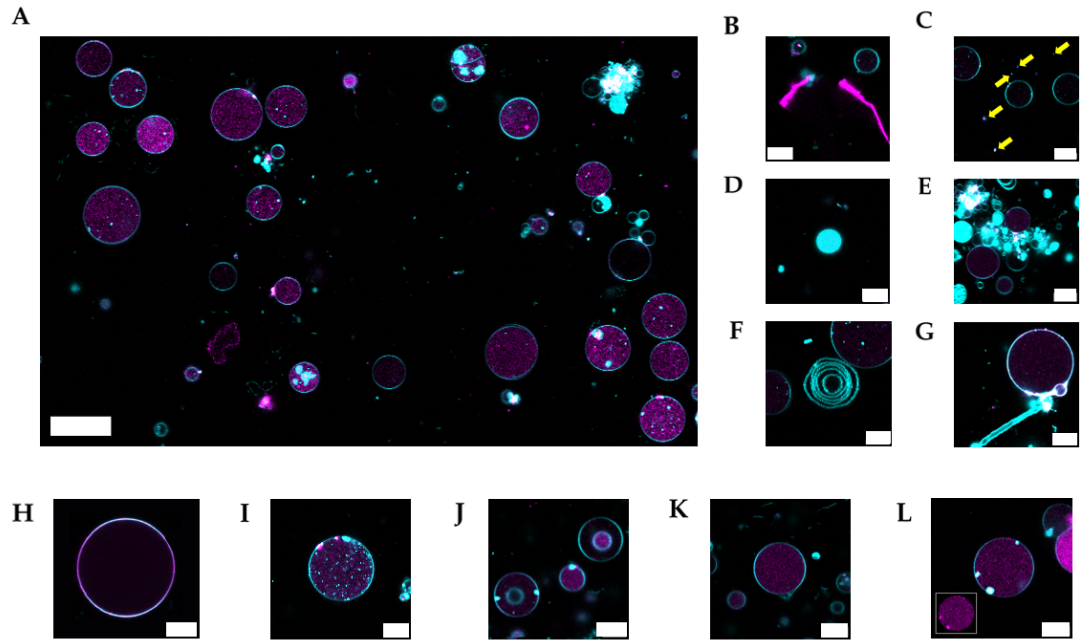


Figure 3.1.: Overview of eDICE sample quality. (A) Typical widefield region of a GUV sample prepared with eDICE. Overlay of the lipid (cyan) and the septin channel (magenta). The GUV membrane composition is 79.75:20:0.2:0.05 DOPC:DOPS:Biotin-PE:Rhodamine-PE and 800nM of septin octamers are encapsulated. Scale bar: 25µm. (B-L) Examples of artifacts and vesicles found in the samples. Different images come from different samples with different membrane compositions and encapsulated content. (B) Un-encapsulated septin bundles or septin aggregation on dust particles. (C) Small lipid aggregates. (D) Oil droplets. (E) Large lipid aggregates attached to vesicles. (F) Multilamellar structures. (G) Lipid tubes. (H) Example image of a GUV displaying no secondary lipid structures. Apart from such GUVs, vesicles encapsulating (I) small free-floating lipids or (J) smaller vesicles were usually present. (K) On occasion, lipid pockets (L) or lipid clusters would be found on the vesicles' membrane. In the latter case septin signal would also be present on those locations, as shown in the inset image. (B-L) Scale bars: 10µm.

3.1. Evaluation of the eDICE GUV samples

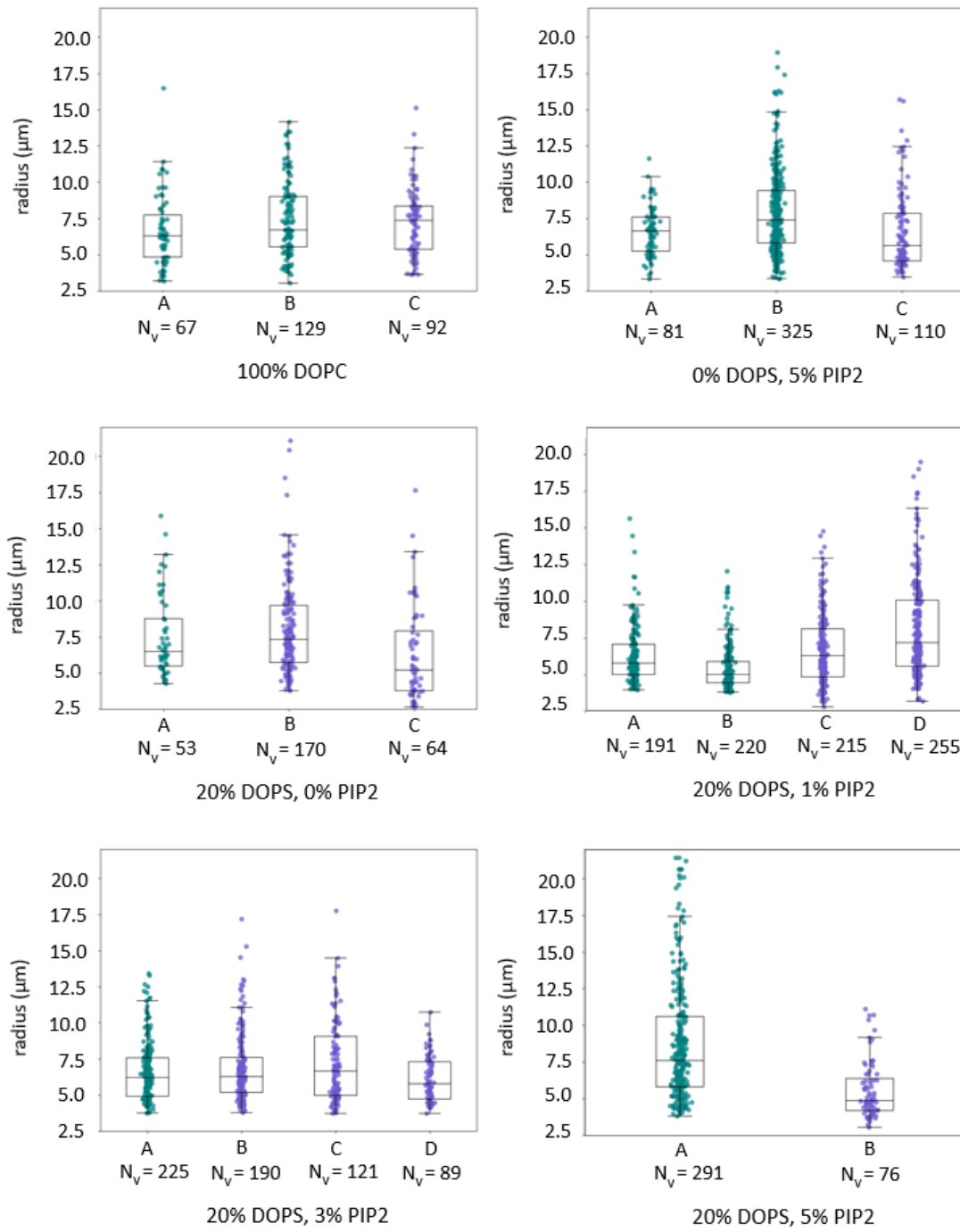


Figure 3.2.: **Radius distributions of GUVs prepared with eDICE.** Individual samples from different days prepared with eDICE (denoted with A-D) and the corresponding numbers of detected vesicles with clean membranes (N_V). Each subfigure corresponds to the denoted membrane composition. Data points corresponding to the radii of vesicles encapsulating septin hexamers (green) or octamers (purple). Only samples that contained more than 50 clean vesicles are presented.

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the present artifacts, after vesicle detection with DisGUVery [77], an automatic filtering was implemented as described in Section 2.5.3 to isolate the clean-looking vesicles and all further statistics correspond to this population.

3.1.2. Septin encapsulation efficiency

For the quantification of the encapsulation efficiency of GUVs produced by eDICE, confocal fluorescence images of 99.75:0.2:0.05 DOPC:Biotin-PE:Rhodamine-PE GUV samples encapsulating septin hexamers at a nominal concentration of 800nM were collected and the pixel intensities of the septin channel were converted to concentrations using calibration measurements of septin bulk solutions (0, 200nM, 400nM, 600nM, 800nM, 1000nM). Salt concentration was kept high (300mM KCl) at all times to avoid septin polymerization. All samples were imaged in the same day, with the same image conditions, and on the same imaging slides, treated in the same way.

Prior to the collection of confocal images of the septin bulk solutions for the calculation of the calibration curve, regions covering approximately 1000x1000 μm were inspected for each sample, to assess the formation of the SLBs at the bottom of the wells and the distribution of the septin signal at the imaging height. A homogeneous lipid signal was present in all the observed regions indicating proper formation of SLBs. The uniform signal was often disrupted by brighter roughly circular patches of diameters between 1-10 μm in all samples (Fig. 3.3A).

The collected images of the septin channel showed the expected homogeneous signal intensity with the exception of the occasional presence of small septin aggregates (1-10 μm) observed as bright specks diffusing around (Fig. 3.3B). However, in the case of the 800nM

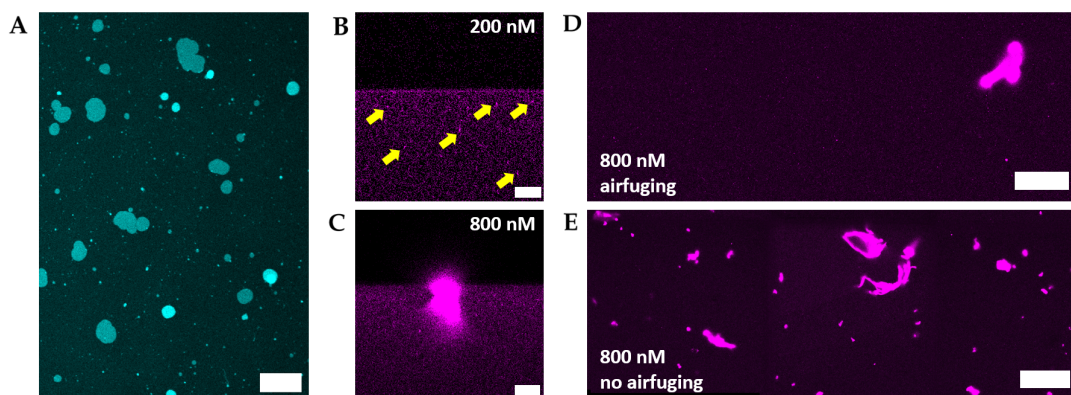


Figure 3.3.: Overview of the septin bulk solution samples. (A) DOPC SLBs were formed on the imaging surface to prevent septin from binding to the glass. Homogeneous formation disrupted by brighter lipid patches was observed. Scale bar: 20 μm . (B-C) xzy-view of the (B) 200nM and (C) 800nM septin bulk solutions. (B) In most regions smaller septin aggregates were present diffusing around (yellow arrows), while (C) larger aggregates were occasionally encountered binding on the imaging surface. Scale bars: 20 μm . (D-E) Septin bulk solutions of 800nM, prepared (D) after airfuging or (E) without airfuging of the septin stocks. As in the first case the presence of aggregates was significantly reduced, all septin bulk solutions used for the calibration were prepared in this manner. Scale bars: 50 μm .

and 1000nM bulk solutions, larger aggregates of lengths ranging between 30 and 80 μm were also observed bound to the imaging surfaces (Fig. 3.3C), possibly at locations of surface imperfections. The amount of those septin aggregates was significantly reduced when septin stocks were airfused during the preparation of the solutions (Fig. 3.3D-E), indicating the possibility of pre-formed aggregates in the septin stock tubes.

As such extensive septin aggregates could additionally exist in other sample regions that were not imaged, it is possible that the obtained mean intensity values from the homogeneous regions of the samples correspond to lower concentrations than the nominal ones. Still, the obtained averages plotted as a function of the nominal septin concentrations showed a good linear relation. Least squares fitting, of the form $I_{\text{sept}}(c_{\text{sept}}) = A \cdot c_{\text{sept}}$ yielded a proportionality constant: $A = 0.0018$ ($R^2=0.97$), indicating that septin binding to the imperfections could be proportional to the septin concentration (Fig. 3.4C).

Imaging of the GUV samples, showed overall successful encapsulation (Fig. 3.4B). To avoid any underestimations in the quantification of the encapsulated content, septin signal should

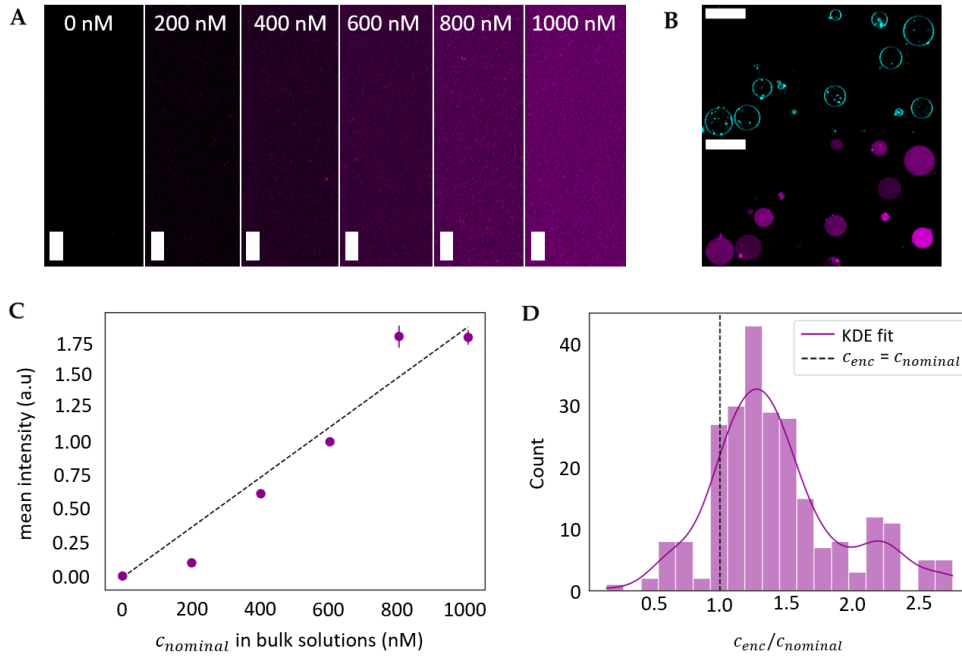


Figure 3.4.: Septin encapsulation efficiency by eDICE. (A) Example regions of each septin bulk solution used for the conversion of pixel intensities to septin concentration. Scale bars: 10 μm . (B) View of the membrane and septin channel at a height of 7 μm above the imaging surface (membrane with cyan and septin with magenta). The membrane composition was 99.75:0.2:0.05 DOPC:Biotin-PE:Rhodamine-PE. The septin concentration in the IAS was 800nM and septin polymerization was inhibited due to the presence of 300nM KCl. Scale bar: 20 μm . (C) Calibration curve (dashed line) obtained after linear fit of the average pixel intensities per septin bulk solution versus the corresponding nominal septin concentrations (magenta points), yielding a proportionality constant $A = 0.0018$ ($R^2=0.97$). Standard deviations of the mean intensities not displayed are smaller than the point size. (D) Histogram and corresponding kernel density estimation (KDE) fit of the encapsulated efficiency after calibration for a nominal concentration of 800nM. The dashed line indicates the nominal concentration. $N_{\text{vesicles}} = 244$.

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be homogeneous throughout the vesicle volume. Therefore vesicles that presented septin aggregation on deformed regions of the vesicle membrane were manually rejected, together with any false detections, leaving 244 GUVs. The distribution of the encapsulated signal was then calculated after averaging within the defined ROIs (Fig. 2.1C) and converting pixel intensities to concentrations using the obtained proportionality constant (Fig. 3.4D).

Compared to the septin concentration of the prepared IAS (800nM), the peak of the obtained distribution appears to be shifted towards higher values (KDE fit peak: 1019nM median: 1088nM, mean: 1133nM) contrary to expectations. This could be a result of the possible underestimation of the proportionality constant A due to the presence of the septin aggregates in the bulk solutions, as the homogeneous regions that were averaged for the calibration curve could potentially correspond to lower concentrations than the nominal ones. The distribution calculated is wide with approximately 14% of the vesicles encapsulating more than double the nominal concentration. Additionally, the distribution seems to be asymmetrical, with a higher percentage of vesicles encapsulating septin concentrations higher than the peak's position (63%), compared to vesicles of low septin content. A similar behaviour has been previously observed for actin encapsulation with eDICE [54], indicating that this apparent supersaturation with protein in a proportion of the vesicles is a characteristic trait of this GUV production method.

3.2. Quantification of septin localization

For a septin meshwork to potentially form inside GUVs, it is necessary that septins can interact and bind to the membrane. Therefore, an investigation of GUVs with different membrane compositions (Fig. 3.5) encapsulating 800nM of septin hexamers or octamers was carried out to determine which lipids can facilitate septin localization on the vesicle membrane.

3.2.1. Lipid specificity of septin binding

DOPC membranes

As expected, vesicles of neutral DOPC membranes mainly demonstrated a homogeneous septin signal within the vesicle lumen, showing no signs of septin localization on the membrane (Fig. 3.6A). In a significant number of vesicles, septin could be found clustered in segments of the vesicle membrane where bright lipid clusters were also visible, possibly indicating membrane deformations and secondary structures (Fig. 3.6D). As in these cases no additional homogeneous localization could be observed along the rest of the membrane, it is most likely that septin is physically trapped in these regions rather than actually binding to the lipids. z-stacks additionally confirmed that this is the case for the whole vesicle membrane and also showed that such clusters are frequently encountered for most vesicles near the imaging surface, indicating that some of the clusters can also be external from accumulation of smaller vesicles around the GUVs (Fig. 3.6D, first vs last panel). The clusters were often visible in the radial profile as increased septin signal on the membrane's region that could falsely be interpreted as localization, and in the angular profile as variations in the corresponding angles (see Appendix Fig. A.2). Therefore, to avoid false classification of these vesicles in the implemented quantification process, vesicles that demonstrated high

3.2. Quantification of septin localization

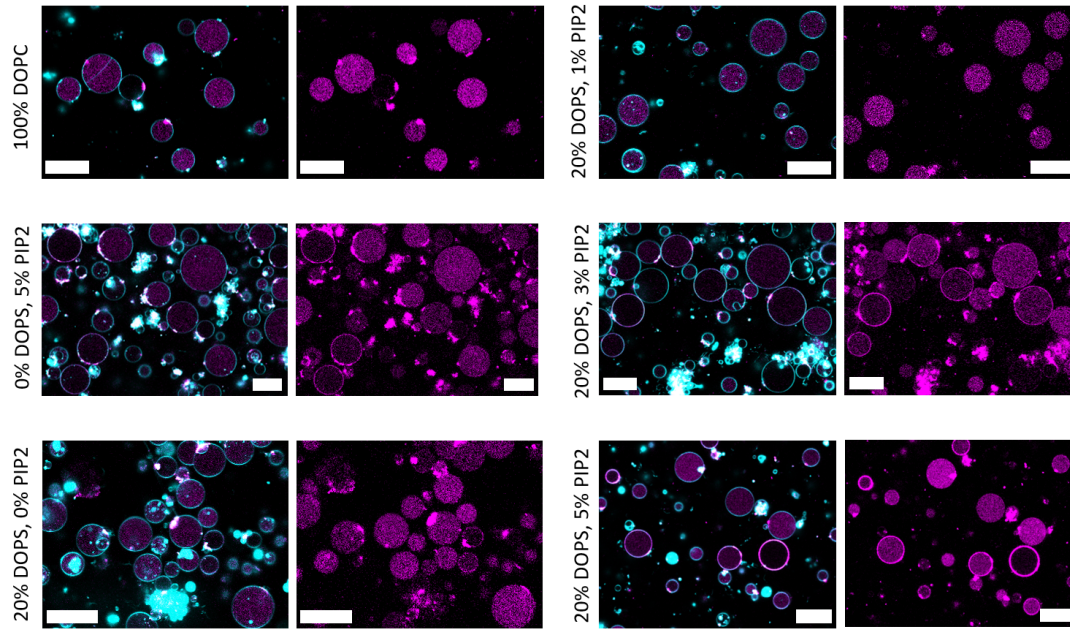


Figure 3.5.: **Overviews of GUV samples encapsulating septin.** Overlay of the membrane (cyan) and the septin (magenta) channels (left), together with the corresponding isolated septin channel (right) for the six investigated membrane compositions denoted in the image. All regions correspond to septin hexamer encapsulation (800nM). Septin localization on the membrane can be seen for some vesicles found in the 0% DOPS 5% PIP2; 20% DOPS 3% PIP2 and 20% DOPS 5% PIP2 samples. Scale bars: 25 μm .

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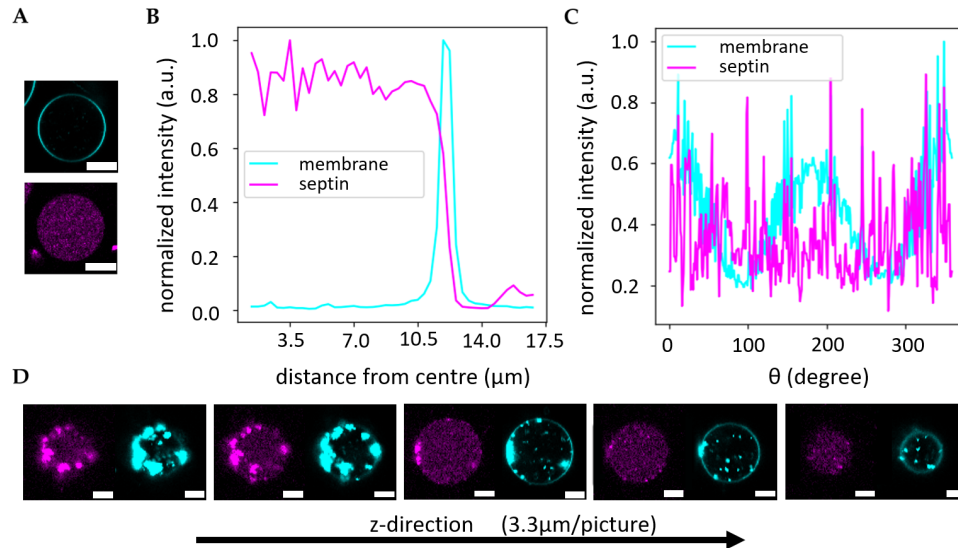


Figure 3.6.: **Septin distribution inside 100% DOPC GUVs.** (A) Example image of a 100% DOPC vesicle encapsulating septin octamers, showing no septin localization on the membrane. Scale bars: 10 μm . (B) The vesicle's radial and (C) angular profiles. (D) z-stack of a 100% DOPC GUV displaying no septin localization in the equatorial plane. Septin signal remains uniform for all heights, with localized clusters only for membrane regions that display similar lipid clustering. The amount of such lipid secondary structures was usually higher near the imaging surface (left) than in higher planes (right). Scale bars: 5 μm

relative standard deviations (RSD) in their angular profiles were removed from the analysis (See methods 2.5.3, last condition). Additionally, as the threshold value of the RSD of the angular profiles could not be too low because of high septin noise, to avoid overestimation of the septin signal on the membrane in the case of undetected smaller clusters, the median of the angular profile was considered in the quantification of septin localization instead of the mean.

Negatively charged membranes

Following the observations of previous studies of fly septins [36], GUVs of DOPS and PIP₂-containing membranes were additionally studied. As these lipids are negatively charged, it was, thus, possible to assess whether electrostatic interactions are similarly important in membrane binding for human septins and whether any lipid-specific interactions exist. To this end, vesicles with only one type of anionic lipids were prepared, either with 80:20 DOPC:DOPS or 95:5 DOPC:PIP₂, as well as samples of 20% DOPS and gradual increase of PIP₂ content. In all cases the rest of the membrane was comprised of DOPC with a trace amount of Rhodamine-PE for fluorescent labeling and Biotin-PE to bind the GUVs to the surface.

In contrast to the DOPC GUV samples, along with vesicles presenting homogeneous septin signal in their lumen, vesicles with homogeneous enriched septin signal along the membrane (Fig. 3.7A) could also be found for all the different samples of negatively charged membranes. z-stacks of such vesicles showed that septin seems to distribute homogeneously

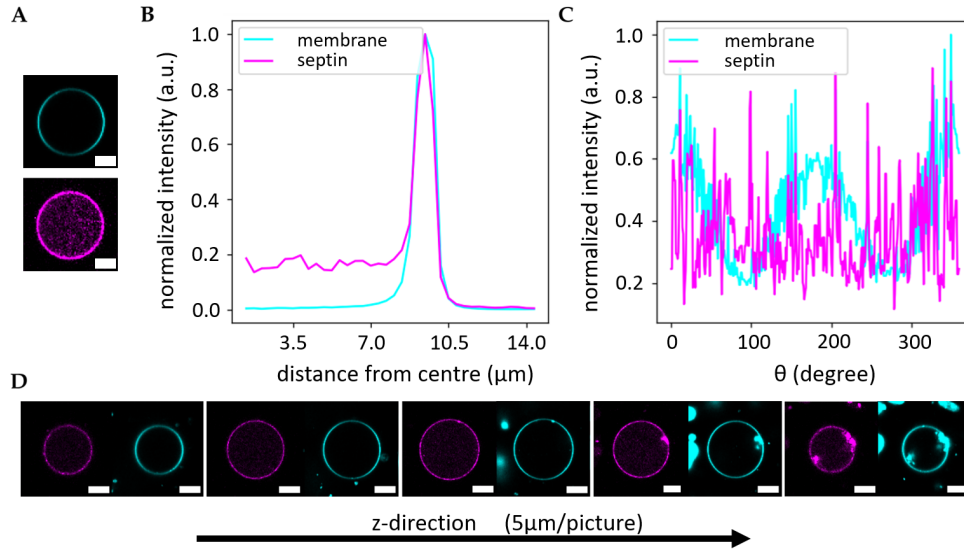


Figure 3.7.: **Septin localization inside GUVs containing negatively charged lipids.** (A) Example image of a 75:20:5 DOPC:DOPS:PIP2 vesicle encapsulating septin octamers, showing septin localization on the membrane. Scale bars: 5 μm (B) The vesicle's radial and (C) angular profiles. (D) z-stack of a GUV of the same membrane composition displaying septin localization in the equatorial plane. Septin localizes homogeneously on the membrane for all heights, with occasional localized septin clusters in membrane regions that display similar lipid clustering. Scale bars: 10 μm

throughout the whole membrane (Fig. 3.7D). Quite often, clusters of septin would also be observed for specific membrane segments at locations of membrane clusters as in the case of the DOPC samples. However, vesicles would still present a homogeneous septin distribution along the vesicle circumference for different angles of the 2D views and along the observed z-stack (Fig. 3.7D, last two panels). As previously, samples with increased variance in the angular profiles were rejected to avoid false classifications and the median septin signal at the membrane was considered.

Even from visual inspection, it was evident that the proportion of vesicles with localized septin varied per membrane composition studied and, thus, different membrane compositions corresponded to different binding affinities. The large number of prepared samples and the implementation of automatic rejection of undesired detected objects allowed for the statistical analysis of the septin localization. The resulting classification and localization distributions are presented in Fig. 3.8 and 3.9 and discussed in the following subsections.

3.2.2. Differences in hexamer and octamer localization

Both common traits and differences were observed in the lipid specificity of the two septin oligomers. For both hexamers and octamers, no septin localization was evident in the case of the neutral DOPC membranes, while septin localization was observed for a subset of vesicles in all samples with negatively charged membranes. Furthermore, in both cases, septin localization was higher in the presence of both DOPS and PIP2 compared to membranes containing only one of those lipids, indicating a synergetic effect in facilitating septin localization.

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However, while in the case of hexamers we could observe higher percentages of vesicles with septin localization only when PIP2 was present in the membrane, indicating selectivity of membrane binding for PIP2, octamers showed high localization rates for membranes with 80:20 DOPC:DOPS as well. Furthermore, in the case of the hexamers a gradual increase of the PIP2 content in presence of 20% DOPS led to a gradual increase of septin localization. In the case of octamers, even the addition of a small PIP2 content in membranes with 20% DOPS seemed to lead to a quite high percentage of septin localization that is maintained despite further increase of the PIP2 (47-51%). The described results for membranes containing both DOPS and PIP2 are also reflected in the medians (Table 3.1) of the corresponding distributions of the quantified localization (Fig. 3.9, marked with red).

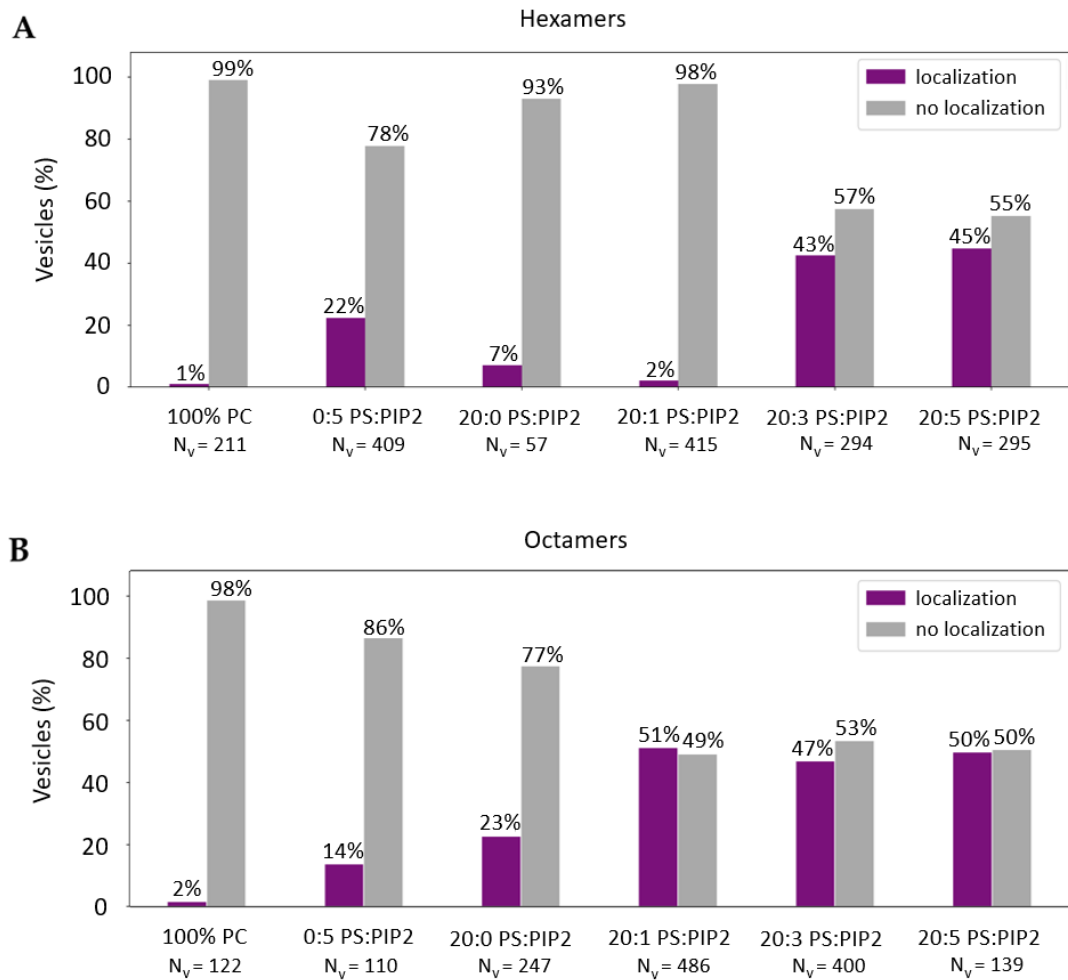


Figure 3.8.: Classification of GUVs based on septin spatial distribution inside the vesicle lumen. Percentages of GUVs with (purple) or without (grey) septin localization on the membrane for all investigated membrane compositions in the case of 800nM of (A) encapsulated septin hexamers and (B) encapsulated septin octamers.

3.2. Quantification of septin localization

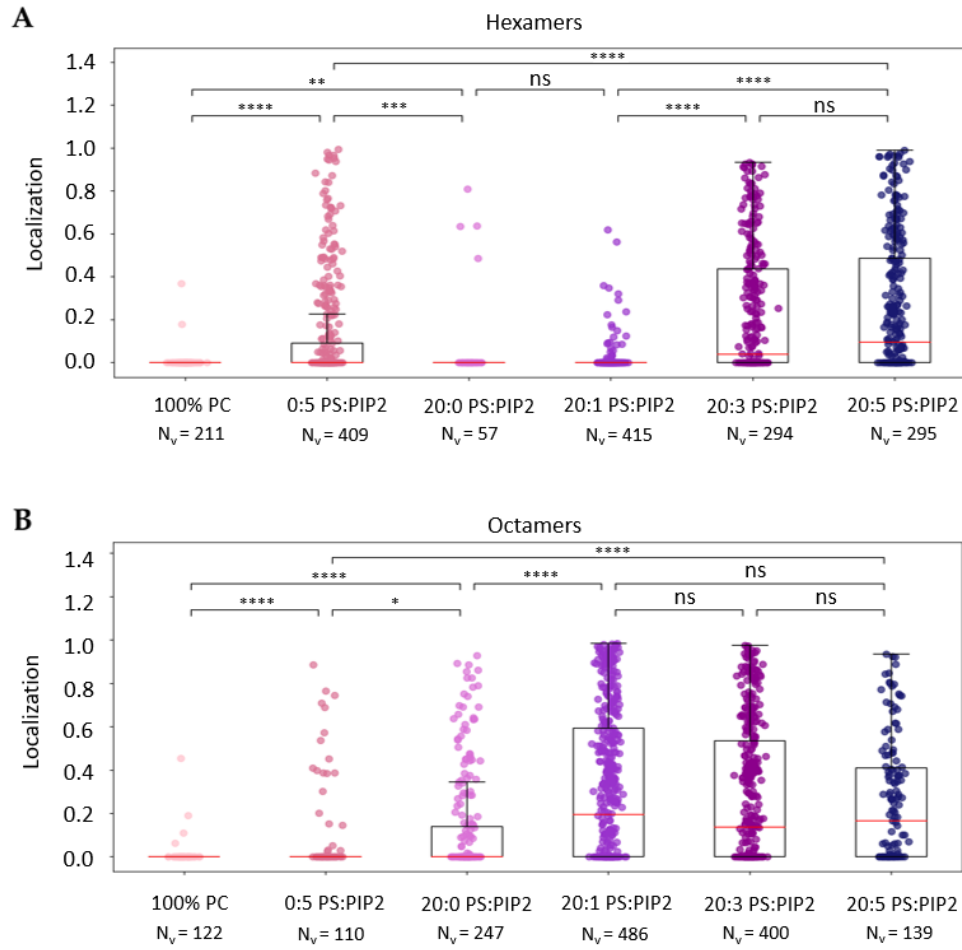


Figure 3.9.: **Distributions of the quantified septin localization values.** Localization values obtained from (Eq. 2.1) for all investigated membrane compositions in the case of (A) encapsulated septin hexamers and (B) encapsulated septin octamers. Negative localization values have been clipped to 0. Individual points indicate single GUVs and boxplots indicate quartiles and medians (red). The p-values are displayed in Appendix Table A.3.

Median values						
<i>PC:PS:PIP2</i>	100:0:0	95:0:5	80:20:0	79:20:1	77:20:3	75:20:5
Hexamers	0	0	0	0	0.038	0.096
Octamers	0	0	0	0.196	0.137	0.165

Table 3.1.: Median values of the quantified localization distributions for different membrane compositions, in terms of molar ratios, and type of encapsulated septin oligomer.

3.2.3. Variability per sample and with membrane curvature

Comparing the percentages of vesicles with membrane-localized septin between different samples of the same membrane compositions, for samples that contained at least 25 clean vesicles, variations ranging from 1% to approximately 60% are observed (Fig. 3.10A-B). The wide range in variations is to be expected as the number of clean vesicles found in each sample differs significantly affecting the percentages. Additionally, variations could result from possible differences in the incorporation of DOPS and PIP2 in the membrane of the vesicles between prepared samples and/or due to the different amounts of encapsulated content (see Section 3.1.2). In general, variations per sample seem to be larger for the membrane compositions that include PIP2, suggesting a higher sensitivity of septin recruitment to PIP2 compared to DOPS for both oligomers.

As septins are able to sense and induce curvature in the micron-scale, the percentage of vesicles displaying localized septin was reviewed in relation to the vesicle size, in order to determine if any preferences in radii, and by extension membrane curvature, are visible from our samples. For the samples where localization was present, there were vesicles with localized signal for all produced radii for both oligomers. Focusing on the range of approximately 4 to 9 μm radius, corresponding to the majority of the vesicles, with a binsize of 0.5 μm , the amount of vesicles with localized septin found in every bin was approximately constant and amounted to 40-60% of the total number of vesicles in each condition (Fig. 3.10C-F). Outside this radius range the limited number of vesicles does not allow any reliable analysis. Therefore, no correlation between localization and vesicle size could be observed from our data.

3.3. Vesicle fluctuation analysis

To assess the effect of septin localization on membrane mechanics, GUVs encapsulating septin hexamers or octamers were prepared with membranes containing 20% DOPS and 5% PIP2, as this composition was previously found to lead to the highest percentage of vesicles with signs of a septin localization. For the quantification of the bending rigidity and the membrane tension, time-lapse movies were collected for the analysis of the observed membrane fluctuations.

3.3.1. Limitations from the sample quality and the imaging setup

To be able to confidently determine whether any observed changes in the membrane's behaviour originate from the presence of membrane-localized septin, septin fluorescence should also be monitored along the lipid signal and a control sample of vesicles known to have no septin bound to the membrane should be studied in parallel to the system of interest. Preparation of 20% DOPS and 5% PIP2 vesicles without encapsulated septin for control measurements consistently resulted in a high number of small lipid aggregates in the vesicle lumen (Fig. 3.11A). The increased presence of such artifacts, apart from possibly inflicting mechanical influences, severely interfered with the precision of membrane detection. In combination with the other secondary structures commonly encountered in eDICE samples (see Section 3.1.1), the collection of time-lapse movies of clean membranes was thus unsuccessful. As the high presence of bright lipid specks could possibly be a result of the

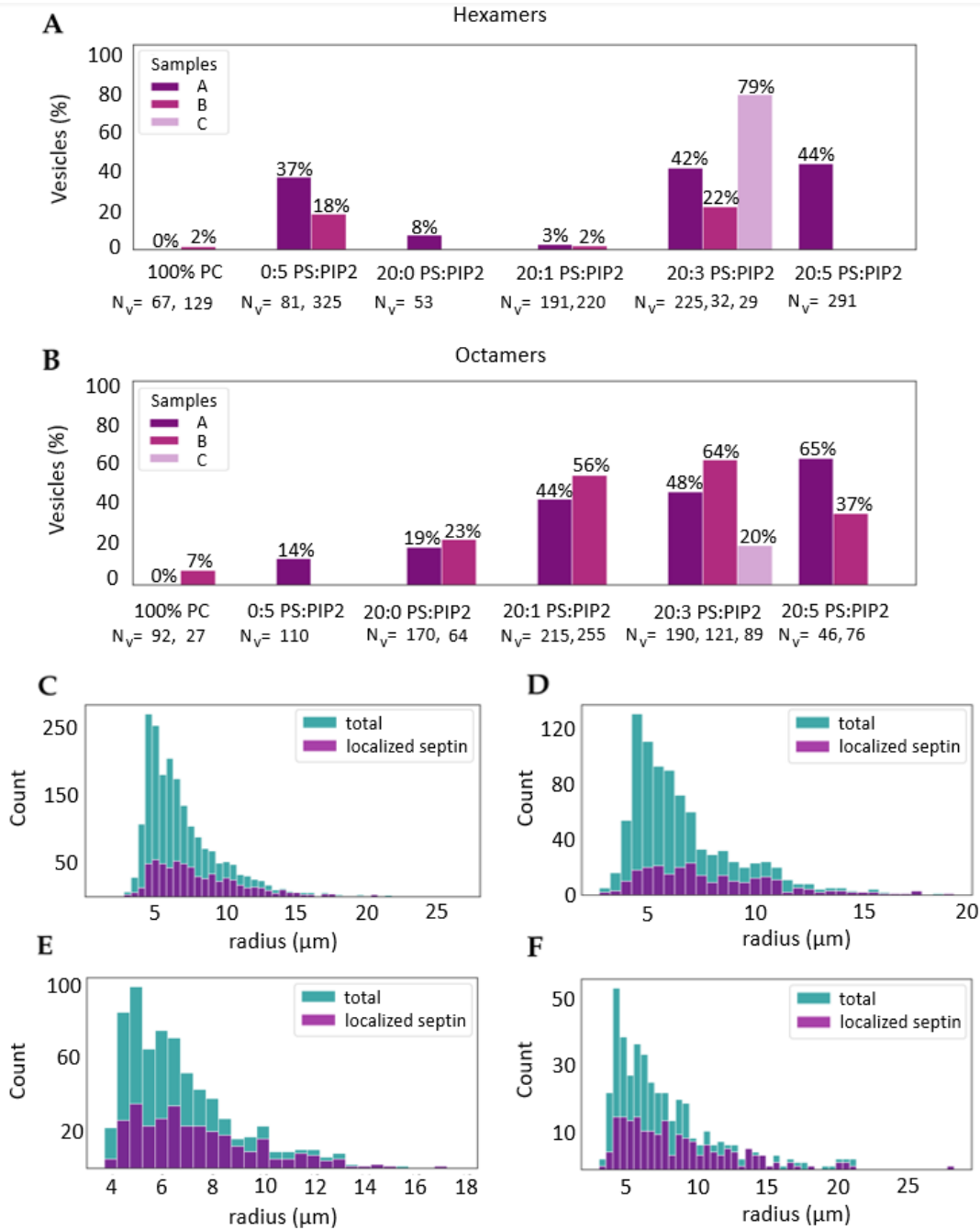


Figure 3.10.: Variability in septin localization per sample and with membrane curvature. (A-B) Variability in the percentage of vesicles displaying septin localization between different GUV samples of the same membrane compositions encapsulating (A) septin hexamers and (B) septin octamers. Only samples containing more than 25 vesicles each are included. (C) Accumulated radius distribution of all the vesicles of compositions 79:20:1, 77:20:3 and 75:20:5 PC:PS:PIP2 (cyan) along with the radius distributions of vesicles of these compositions with localized septin (magenta). (D-F) Radius distributions of all the vesicles of a specific membrane composition (cyan) along with the radius distributions of vesicles with membrane-localized septin (magenta). Displayed compositions: (D) 20% DOPS, 1% PIP2, (E) 20% DOPS, 3% PIP2, (F) 20% DOPS, 5% PIP2. All binsizes are $0.5\mu\text{m}$.

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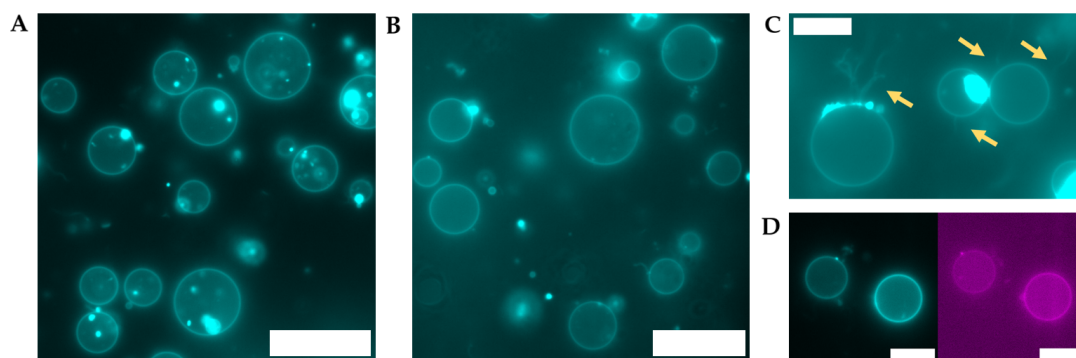


Figure 3.11.: Overview of epifluorescence images of 75:20:5 DOPC:DOPS:PIP2 and 100% DOPC GUV samples. (A) Typical widefield region of empty GUVs of 20% DOPS, 5% PIP2 membranes. Bright lipid aggregates diffusing in the lumen were observed in the vast majority of the vesicles, severely interfering with membrane detection. Scale bar: 20 μm . (B) Typical widefield region of 20% DOPS, 5% PIP2 GUVs encapsulating septin hexamers. The number of encapsulated lipid aggregates was reduced compared to the empty samples. Scale bars: 20 μm . (C) Lipid tubes (yellow arrows) were often observed in the majority of the vesicles. Scale bars: 10 μm . (D) Crosstalk between the membrane (cyan) and the GFP-labelled septin (magenta) channel was present even when Rhodamine-PE was replaced by ATTO 655 DOPE, not allowing discrimination between vesicles with and without localized septin. Scale bars: 5 μm .

high net negative charge of the membrane or of poor solubility of PIP2, DOPC vesicles were prepared instead to be used as a point of comparison. Additionally, as vesicles of 20% DOPS and 5% PIP2 encapsulating septin consistently showed less such artifacts in the vesicle lumen (Fig. 3.11B), control vesicles were also prepared encapsulating septin octamers in an attempt to limit the specks occurrence. As DOPC membranes do not facilitate septin binding (see Section 3.2.1), the septin content should not affect the mechanical properties of the membranes.

An additional hurdle encountered was the presence of lipid tubes in the vast majority of vesicles (Fig. 3.11C). Such structures possibly form from excess membrane area generated from the high osmolarity difference between the inside and the outside of the vesicles. As excess membrane area in these cases is stored in such lipid extensions, such vesicles appeared rigid-like presenting no visible fluctuations. The occurrence of this phenomenon was similar for all prepared samples of different membrane compositions. In an attempt to limit the extent of this phenomenon, vesicles were initially prepared with a low osmotic difference between the IAS and the outer buffer (approximately 15 mOsm/kg) and left incubating for around 15 minutes prior to dilution in the imaging well and gradual increase of the osmolarity through evaporation by lifting the imaging chambers lid for 15-20 minutes. While evaporation led to an increased number of fluctuating vesicles over time, lipid tubes were always widely present in the samples, limiting the number of available vesicles for the fluctuation analysis.

A final significant limitation in the interpretation of the results was the high crosstalk levels between the septin and the lipid signal in the imaging setup used to acquire the fluctuation data. In combination with low septin signal, this resulted in vesicle membranes being visible in the septin channel, limiting our ability to identify vesicles with actual septin localization on the membrane. Modifications of the LED intensity and use of the available filters did

not successfully resolve this issue. As GFP-labeled septins have an emission wavelength of 488nm and Rhodamine-PE lipids have an emission peak at 568nm, the alternative dye-lipid ATTO 655 PE was also used, to limit fluorescent crossover. However, as crosstalk was still evident in those samples (Fig. 3.11D), we reverted to the use of Rhodamine-PE. As we were unable to discern between vesicles with and without septin localization, and as we investigate the possibility of membrane rigidification due to septin binding, data collection was focused on clean vesicles observed in the samples under study and the control system, despite the presence of visible fluctuations or a seemingly rigid membrane contour.

3.3.2. Bending rigidity and membrane tension measurements

Initial inspection of the prepared samples revealed a large number of rigid vesicles in all samples, commonly exhibiting secondary lipid structures as described previously. Fluctuating vesicles would also be observed, mostly with higher frequency in the case of the control samples and for vesicles of smaller sizes (3-5 μm radii). After the removal of the imaging chamber lid to allow evaporation, larger vesicles could gradually also be seen to fluctuate. Fluctuating vesicles could also be observed in the 20% DOPS, 5% PIP2 samples where septin localization is expected in approximately 50% of the present vesicles. Still, the presence of rigid vesicles was quite high, comprising between 36-60% of the observed clean vesicles in the control samples and 40-75% in the samples with negatively charged membranes (see Appendix Table A.4).

Rejection of the vesicles with problematic membrane detection, unclear intermediate mode regimes, and inconsistent bending rigidity values for different choices of the lower cutoff mode for the fitting, led to the isolation of the vesicles of quantifiable bending rigidity and membrane tension. An example of successful quantification for a vesicle of membrane composition containing 20% DOPS, 5% PIP2, and encapsulating septin octamers is displayed in Fig. 2.3.

During image acquisition, high LED intensity values (50%) were required for a sufficient contrast to facilitate membrane detection. Additionally, due to variations in background noise, time-lapse videos of individual vesicles were collected with different exposure times, τ , varying from 5 to 50 ms for improved contrast. To determine whether consistent values of the bending rigidity can be obtained within this range of exposure times, time-lapse videos of the same vesicle were obtained for varying τ . Due to the high intensities leading to photobleaching and the occasional appearance of bright specks interfering with proper tracking of the membrane contour not all videos with different exposure times led to successful quantification for each vesicle. The obtained bending rigidity values, κ , for the successful subset of these attempts are presented in Fig. 3.12A. Consistent κ values are obtained for such variations of the exposure time in agreement with previous observations [79], allowing for reliable comparisons between the bending rigidities of different vesicles obtained with exposure times within this regime.

An overview of all fitting results for all samples and the corresponding vesicle radius and exposure time is displayed in Table 3.2. Quantitative analysis of the fluctuating vesicles led to membrane tension values in the range of 10^{-10} - 10^{-7} N/m in all samples. These values are within the same range as in previous VFA experiments carried out with the same imaging setup [79] and as VFA measurements of biomimetic membranes [69, 81]. Vesicles of higher tensions could not be analyzed, as with our approach the lower modes of the fluctuation spectrum cannot be optically resolved. An investigation by Lennard van Buren

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demonstrated that σ in the range of 10^{-10} - 10^{-7} N/m has a minimal effect on the fluctuation spectrum and, therefore, apart from the order of magnitude, the obtained values are not a reliable measure of the membrane tension [79].

The obtained distributions of κ values for each condition from the quantitative fluctuation analysis are plotted in Fig. 3.12A. Vesicles of DOPC membranes encapsulating septin octamers, for which no septin localization is expected, were found to consistently have κ values from 8 to 14 $k_B T$. In contrast, measurements of GUVs with membranes containing 20% DOPS and 5% PIP2 encapsulating either hexamers or octamers led to a wider distribution of κ values, ranging from 10 to 35 $k_B T$. As described in Section 3.2, quantification of septin localization with confocal imaging showed that approximately 50% of the vesicles with 75:20:5 DOPC:DOPS:PIP2 membranes display septin localization on the membrane (Section 3.2), while additionally quantification of septin encapsulation efficiency showed that the encapsulated septin concentrations can vary significantly (Section 3.1.2). Therefore, in the DOPS and PIP2 containing samples, we expect to encounter both GUVs with localized septin and vesicles without septin localization due to low or no encapsulated septin content. If septin is indeed increasing the membrane rigidity through septin-membrane interactions, this would lead to two distinct populations of bending rigidities around a higher and a lower value, the latter comparable to the κ values found in the control system, where no localization takes place. As we do observe a wider distribution of measured κ values in the DOPS and PIP2 containing samples and, especially in the case of septin octamers, we can observe signs of two distinct populations of vesicles of a higher (~ 30 $k_B T$) and a lower (~ 15 $k_B T$) bending rigidity, it is very likely that the observed increase in bending rigidity is a result of septin-membrane interactions.

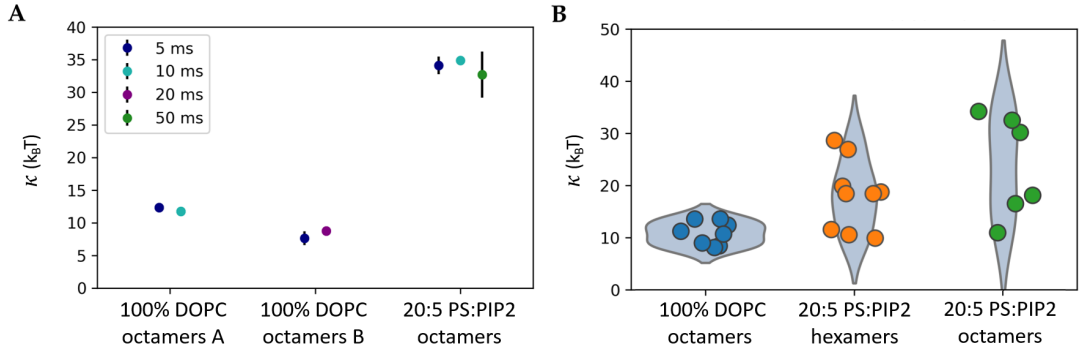


Figure 3.12.: Values of bending rigidity obtained with VFA. (A) Bending rigidity values for three different vesicles of the denoted membrane compositions from time-lapse videos taken with different exposure times. Photobleaching and the occasional appearance of bright fluorescent specks did not allow quantification for all tested exposure times for each single vesicle. Errorbars not depicted are smaller than the marker size. (B) Bending rigidity values obtained for the control system (blue, $N_{\text{vesicles}}=8$) and hexamer (orange, $N_{\text{vesicles}}=9$) and octamer (green, $N_{\text{vesicles}}=6$) encapsulating vesicles with negatively charged membranes. Points correspond to individual vesicles.

Analyzable data					
Sample	radius (μm)	τ (ms)	σ (N/m)	κ ($\text{k}_\text{B}\text{T}$)	$\delta\kappa$ ($\text{k}_\text{B}\text{T}$)
control	5.5	5	$2.4 \cdot 10^{-9}$	10.62	0.10
control	4.9	5	$1.2 \cdot 10^{-8}$	8.05	0.23
control	5.9	5	$3.8 \cdot 10^{-9}$	12.3	0.3
control	8.3	5	$2.4 \cdot 10^{-9}$	13.51	0.20
control	6.2	10	$1.2 \cdot 10^{-8}$	8.4	0.4
control	4.5	10	$3.8 \cdot 10^{-8}$	13.5	0.3
control	7.6	20	$6.4 \cdot 10^{-8}$	8.9	0.3
control	11.3	50	$8.7 \cdot 10^{-9}$	11.1	0.7
hexamers	7.4	5	$5.6 \cdot 10^{-7}$	19.8	0.8
hexamers	3.8	5	$3.1 \cdot 10^{-9}$	11.49	0.13
hexamers	6.9	5	$4.6 \cdot 10^{-8}$	28.6	1.1
hexamers	5.2	5	$1.2 \cdot 10^{-7}$	18.4	1.6
hexamers	9.2	20	$3.3 \cdot 10^{-9}$	26.8	1.0
hexamers	7.4	50	$6.0 \cdot 10^{-10}$	10.49	0.18
hexamers	6.9	50	$3.3 \cdot 10^{-8}$	18.7	0.9
hexamers	4.3	50	$1.3 \cdot 10^{-8}$	18.4	1.1
hexamers	7.7	50	$4.5 \cdot 10^{-9}$	9.8	0.6
octamers	9.2	5	$2.8 \cdot 10^{-9}$	34.1	1.3
octamers	5.8	5	$2.9 \cdot 10^{-8}$	30.1	0.5
octamers	4.3	10	$8.0 \cdot 10^{-9}$	10.9	1.0
octamers	8.3	10	$4.4 \cdot 10^{-11}$	32.4	1.0
octamers	5.9	10	$7.7 \cdot 10^{-9}$	18.0	0.6
octamers	7.3	20	$6.7 \cdot 10^{-9}$	16.4	0.9

Table 3.2.: Fitting results for κ and σ . Control GUV membranes consisted of 99.7:0.2:0.1 DOPC:Biotin-PE:Rhodamine-PE and encapsulated septin octamers. The vesicle membranes of GUVs encapsulating either hexamers or octamers consisted of 74.7:20:5:0.2:0.1 DOPC:DOPS:PIP2:Biotin-PE:Rhodamine-PE.

Despite the increase in the amount of fluctuating vesicles after evaporation, rigid-like vesicles were still widely present in samples of all three different conditions. While membrane tracking was still possible in these cases, the utilized script was unable to properly calculate the fluctuation spectrum due to the small magnitude of the membrane undulations. To acquire a sense of the membrane's deformability due to the fluctuations, the root mean square displacement, X_{rms} , between successive frames was measured for a rigid vesicle randomly selected per condition for time-lapse videos all obtained with an exposure time of 5ms. No considerable differences were observed between the rigid vesicles in the control and in the systems under study as the magnitude of X_{rms} was in all cases smaller than the pixel size, indicating limitations due to spatial resolution (see Appendix Fig. A.3).

4. Discussion

4.1. Lipid specificity in human septin-membrane binding

Human septins interact with DOPS and PIP2 membranes

Encapsulation of human septin hexamers and octamers within GUVs of various membrane compositions and quantification of septin localization revealed that human septins can interact with membranes consisting of the negatively charged lipids, DOPS and/or PIP2, in contrast to neutral DOPC membranes where no signs of septin localization are present. This result is in agreement with observations from fly septins [36], as well as previous studies of human octameric complexes [33], and indicates the contribution of electrostatic forces in human septin-membrane interactions. Additionally, the presence of both DOPS and PIP2 was observed to lead to increased percentages of vesicles displaying septin localization compared to membranes incorporating a single type of anionic lipid, implying the presence of a synergistic effect in septin recruitment. This observation is consistent with previous studies of human octameric complexes, where GUVs incubated with septins showed increased interactions in presence of both DOPS and PIP2 compared to membranes lacking PIP2 [33]. Furthermore, we observed that a gradual increase of PIP2 in the membrane led to a significant increase in the percentages of vesicles with localized septin, possibly as a result of lipid-specific interactions.

In our group, a study of human septin binding on SLBs has been carried out by Gerard Castro-Linares with Quartz Crystal Microbalance with Dissipation monitoring (QCM-D), an acoustic technique that detects mass changes at an oscillating sensor through calculation of the resulting shifts in the surface's resonance frequency Δf and dissipation ΔD [82]. After formation of 80:20 DOPC:DOPS SLBs on the sensor's surface, perfusion with septin bulk solutions of 50nM in a buffer of the same composition as the IAS we used, without the inclusion of the density medium (OptiPrep), consistently led to no significant frequency shifts. This indicates that no adsorption took place for either hexamers or octamers. Combining this results with our observations of GUVs of 80:20 DOPC:DOPS membranes where localization is sometimes present, especially in the case of octamers, suggests the possibility that the negative charge of DOPS is sufficient to attract septins close to the membrane but cannot necessarily facilitate (strong) membrane binding. Therefore, no adsorption is observed in the QCM-D experiments as septins are provided with a constant flow that could wash away any proteins weakly interacting with the bilayer. In the case of our GUV samples, it is possible that septin subunit exchange takes place at the membrane where septin localization is observed instead of the formation of a stably membrane-anchored meshwork or that the higher septin concentration encapsulated (800nM) and the consequently increased probability of oligomers coming together eases polymerization, stabilizing membrane binding.

For membrane compositions of 75:20:5 DOPC:DOPS:PIP2, QCM-D clearly showed large mass changes indicating septin adsorption. For the same composition we also observed

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the highest percentages of vesicles with localized septin signal for both oligomers. Thus, it is very likely that such membranes promote strong binding and formation of a dense septin meshwork as observed on solid substrates [33]. The highest localization ratios observed in presence of both 20% DOPS and 5% PIP2 compared to when only 5% PIP2 is present could possibly result from assisted septin recruitment due to the higher membrane net charge. As in parallel septin localization appears sensitive to the presence of PIP2 in the membrane, it is likely that while DOPS helps in increased septin localization in the near proximity of the membrane's surface, stable membrane anchoring is facilitated by specific septin interactions with PIP2.

No observed curvature preference for the investigated negative curvature range

Previous studies on yeast [41, 67] and human septins [33] have demonstrated that septin filaments are both able to sense and induce curvature in the micron-scale. For both yeast and human septins, an increased affinity for membranes with curvatures in the range of $0.7\text{-}2\ \mu\text{m}^{-1}$ has been reported from an investigated range from 0 to $3\ \mu\text{m}^{-1}$. At the same time, membrane-bound septin filaments have been observed to preferably bind to negative curvatures, orientating themselves along the curve, in contrast to positive curvatures where they align parallel to the null curvature axis, avoiding being bent [41, 33]. In the latter case, when septins localize on the outside of free-floating GUVs, they were observed to induce periodic bump-like deformations with negatively curved areas formed in between those hills, indicating the potential of septins to generate their preferred curvatures. This behaviour has been reported by Nakazawa et al.[33] using 10-100nM of human septin octamers, and observed by Gerard Castro-Linares in our group, with 800nM of either human septin hexamers or octamers, with membranes containing PS and PIP2.

In our model system, septins were encapsulated inside GUVs of radii in the range of 3 to $25\ \mu\text{m}$ (Fig. 3.2). Thus, they were in presence of a negatively curved bilayer of curvatures between 0.04 to $0.33\ \mu\text{m}^{-1}$. Septin localization was observed in vesicles of all observed sizes. Our vesicle formation method produces a significant number of vesicles of diameters between 8 to $18\ \mu\text{m}$, corresponding to curvatures of $0.1\text{-}0.25\ \mu\text{m}^{-1}$. Further investigation in this size range for each membrane composition showed approximately constant percentages (40-60%) of vesicles exhibiting septin localization for all radii with a binsize of $0.5\ \mu\text{m}$. Thus, no curvature preference could be observed in our samples. The curvatures encountered in our system are lower than the curvatures previously shown to be sensed by septins even in the case of our smaller vesicles, suggesting the possibility that septins might sense a "flat" surface within our explored range of negative curvatures. Additionally, it is possible that we cannot detect any clear preference due to the high septin concentrations that we encapsulate (800nM). This concentration is higher than the 90nM observed to be sufficient for the formation of a saturated double layered human octameric septin meshwork that covers an undulated solid substrate fully and in a uniform manner [33]. The membranes of our vesicles can similarly be saturated, obscuring the detection of any preferred curvature. Finally, in our samples we do not observe any periodical micro-scale deformations, in contrast to the case when septin is introduced on the outside of GUVs. Thus, we confirm that the curvature sensitivity of septins differs for negative and positive curvatures and in the latter case local membrane re-shaping probably originates from septins' ability to generate areas of negative curvature to which septins preferentially bind [33].

4.2. Different lipid affinity for septin hexamers and octamers

Both septin hexamers and octamers were observed to localize on a subset of vesicles containing either 20% DOPS or 5% PIP2. However, our data indicated different affinities when a single negatively charged phospholipid type is present in the membrane for the two oligomers. Septin hexamers were observed to localize in a significantly higher percentage of vesicles with membranes containing 5% PIP2 (22%) compared to membranes with 20% DOPS (7%). In contrast, GUVs with localized septin octamers would also be frequently observed in samples of GUVs containing 20% DOPS (23%). In addition, gradual increase of the PIP2 membrane content in presence of 20% DOPS led to a gradual increase in the number of vesicles with septin hexamer localization, while in the case of octamers the corresponding percentage significantly increased even after addition of a low amount of PIP2 and remained approximately constant after further PIP2 addition in the membrane.

As the same preparation method was used and all conditions were kept the same for both samples with hexamers and octamers, it is most likely that the observed differences result from structural differences or distinct ways of filament assembly on lipid membranes of the two oligomers. The main structural difference between hexamers and octamers is the presence of the two SEPT9 monomers at the center of the latter. Unlike other septin monomers, SEPT9 does not contain coiled-coils at its C-terminal extension but includes unique N-terminal extensions [74] known to mediate septin-microtubule interactions [83]. The potential structural rearrangements of such domains in presence of a lipid bilayer [74] could perhaps affect septin orientation, assisting in septin polymerization and as a consequence septin-membrane binding stability, in contrast to the SEPT9-lacking hexamers. Furthermore, the presence of SEPT9 differentiates the two oligomers in terms of length and alters the spacing of the coiled-coil domains of the other subunits within a filament. As coiled-coils appear to affect higher order filament organization [36], a difference in their geometrical arrangement within the filaments could potentially affect the condition under which a membrane-bound meshwork can form. Finally, previous studies of membrane-associated septin meshes for human septin hexamers, carried out in our lab by Djim de Ridder, and for human septin octamers [33] report regular inter-filament distances about 21nm and 30nm respectively in agreement with the corresponding oligomer lengths (24nm and 32nm). This difference in the membrane-associated spatial assembly of the two oligomers could possibly be a reflection of different septin-membrane interactions.

4.3. Effect of septin localization on membrane bending rigidity

In order to investigate the effect of a septin meshwork on membrane mechanics, we carried out a quantitative analysis of vesicle fluctuations with epifluorescence microscopy. To ensure that a septin cortex can potentially form, vesicles of 74.7:20:5:0.2:0.1 DOPC:DOPS:PIP2:Biotin-PE:Rhodamine-PE encapsulating either hexamers or octamers were prepared, as higher septin localization rates were observed with this composition for both oligomers. The bending rigidities obtained were compared with the ones of 99:75:0.2:0.05 DOPC:Biotin-PE:Rhodamine-PE vesicles encapsulating septin octamers, where no septin localization is expected. While the majority of the vesicles were seemingly rigid, fluctuating vesicles could be located in both samples, especially after an increase in the osmotic pressure with buffer

4. Discussion

evaporation. Quantitative analysis of the fluctuating vesicles led to membrane tension values in the range of 10^{-10} - 10^{-7} N/m in all samples. Bending rigidities of the control system ranged from 8 to 14 $k_B T$, while in the 20% DOPS, 5% PIP2 samples a wider distribution of values was observed between 10 to 35 $k_B T$ with a maximum estimated error of approximately 1 $k_B T$.

In our group and with the same experimental setup, membrane tension values in the same range were obtained by Lennard van Buren [79] for empty DOPC GUVs produced by gel swelling or electroformation. Additionally, this membrane tension range is in the same order of magnitude with values reported in literature for biomimetic membranes [69, 81]. As the membrane tension dominates in the lower fluctuation modes that cannot be optically resolved with our method [79], the exact obtained values cannot be considered reliable apart from their scale of magnitude. The bending rigidity values measured by Lennard van Buren [79] of DOPC membranes were 16.6 ± 1.8 $k_B T$ for gel-swollen vesicles and 18.6 ± 1.8 for electroformed vesicles, indicating altered mechanical properties based on the production protocol and its related artefacts [79]. The average bending rigidity of the DOPC samples we prepared with eDICE was 10.9 ± 2.2 $k_B T$, lower than the values obtained for the other preparation protocols. The observed difference could potentially originate from the possible presence of oil in the lipid bilayer of our samples, due to the nature of our vesicle preparation protocol, or the different buffers used that could possibly have an effect on the measured rigidity [20].

The wider distributions of κ values observed in the samples expected to facilitate septin localization on the membrane is very likely an indication that septin can increase the rigidity of the membrane through their interactions. From previous results (Section 3.2), it was observed that in samples of such membrane compositions, septin localization takes place in approximately 40-60 % of vesicles. From the quantification of the encapsulation efficiency (Section 3.1.2), we additionally expect that the encapsulated concentrations can vary significantly. Therefore, if septin can indeed rigidify the membrane we would expect the presence of two distinct populations around a higher κ value when septin localizes and potentially a meshwork is formed, and a lower κ value, comparable to the one of the control system and corresponding to no septin localization. Our collected data seem to present a pattern in agreement with this expectation. Additionally, the considerable presence of rigid-looking vesicles in those samples, further illustrates the possibility of membrane rigidification due to septin localization.

Due to our inability to confirm the presence of localized septin on the vesicle's membrane because of high crosstalk levels, in addition to the limited quantifiable data collected due to the high presence of lipid aggregates and tubes, it is not clear whether the increased bending rigidities observed originate solely from membrane interactions with septins. Considering that the presence of lipid tubes and protrusions was very common in all our samples, such undetected structures could potentially contribute to the high numbers of rigid vesicles observed, as excess membrane area could be stored in these structures, limiting fluctuations [84]. This could also explain the presence of high numbers of rigid-like vesicles in the control system. Furthermore, the presence of DOPS and PIP2 could potentially play a role in the observed difference compared to the control system of DOPC membranes. As these lipids are negatively charged, repulsions in the bilayer could potentially lead to a higher bending rigidity [20]. However, as previous studies of tether-pulling experiments with PC membranes with additional 16% PS [85] or 5% PIP2 [86] did not report any detected differences in bending rigidity values compared to pure PC membranes, it is likely that in our system such contributions are not that significant for such molar ratios. Still, further

4.3. Effect of septin localization on membrane bending rigidity

investigation is required for more concrete conclusions regarding the effect of septins on membrane mechanics.

5. Conclusion and outlook

5.1. Conclusion

Prompted by previous research that suggests a dynamic role of septin in the mechanical integrity of cells [4, 27, 48] and its ability to form stiff and ductile membrane-associated meshworks *in vitro* [33, 36], we set out to investigate the possibility of a septin cortical network directly enhancing the rigidity of the cell cortex. To this end, we studied human septin and membrane interactions towards the reconstitution of a septin cortex inside free-floating membranes by encapsulating human septin hexamers and octamers in GUVs.

We quantified the septin encapsulation efficiency by collecting confocal fluorescence images of our samples and converting intensities to concentrations using calibration measurements on septin bulk solutions. We observed that eDICE produces a high yield of GUVs of consistent radii distributions that can robustly encapsulate septin. We found a wide distribution of encapsulated concentrations and observed that a significant percentage of the GUVs was supersaturated with septin, encapsulating higher concentrations than the nominal septin concentration in the IAS. Although, overestimation is likely due to the presence of septin aggregates in the bulk solutions for the calibration measurements, a similar behaviour has also been observed for actin encapsulation with eDICE [54].

To identify human septin-lipid affinity in prospect of reconstituting a membrane-anchored septin network, we quantified septin localization from confocal fluorescence images of GUVs of different membrane compositions containing DOPC and the negatively charged lipids DOPS and PIP2 in different ratios. We observed septin localization in a subset of vesicles in all samples with negatively charged lipids in contrast to neutral 100% DOPC membranes, confirming that electrostatic forces are important in human septin-membrane interactions [33]. Additionally, starting with membranes consisting of 80:20 DOPC:DOPS and gradually increasing the PIP2 content from 0 to 5%, we observed a higher percentage of vesicles with localized septins in presence of both DOPS and PIP2 than in the samples where only one of those lipids was present, indicating a synergistic effect in septin recruitment, possibly due to increased membrane net charge and high septin sensitivity to the PIP2 membrane content. Comparing with QCM-D experiments carried out in our group by Gerard Castro-Linares, we hypothesize that DOPS is sufficient to attract septins close to the membrane but cannot necessarily facilitate strong membrane binding, while PIP2 can lead to strong membrane anchoring through specific interactions.

Furthermore, we identified a different behaviour between septin hexamers and octamers in presence of 80:20 DOPC:DOPS membranes. For this membrane composition, GUVs encapsulating hexamers showed very limited localization compared to 95:5 DOPC:PIP2 membranes, while in the case of the octamers a higher percentage of DOPS containing vesicles was observed to facilitate septin localization comparable to the case of PIP2. Additionally, while septin localization in the hexamer samples increased more gradually with the increase of PIP2, a more sudden increase was observed for the octameric complexes. It is possible

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that the difference in the observed behaviour originates from structural differences of the two oligomers due to the presence of SEPT9 that might lead to distinct ways of filament assembly on lipid membranes but further research is necessary to clarify the molecular mechanisms involved. In our samples, we could not detect any septin preference in curvature for either oligomer, possibly due to the high septin concentrations we use that could saturate the membrane and due to the accessible range of the GUV sizes. With encapsulated septins we observed no periodic deformations, in contrast to the case when yeast septin [41] or human septin hexamers (Gerard Castro-Linares in our group) and octamers [33] are introduced outside of GUVs, suggesting that septin curvature sensitivity differs for negative and positive curvatures.

Finally, to gain insight into the role septin plays in cell cortex functionality, we studied the effect of septin-membrane interactions on the bending rigidity of the vesicle membrane with Vesicle fluctuation analysis using epifluorescence microscopy. We were able to observe a wider distribution with a higher average for the bending rigidity values in samples with membrane compositions 75:20:5 DOPC:DOPS:PIP2 encapsulating septin hexamers and octamers, that can facilitate septin localization, compared to vesicles of 100% DOPC membranes encapsulating septin octamers, where no septin-membrane interactions are expected. It is very likely that the higher values of the bending rigidity are a result of septin localization on the vesicle membranes. While our observations indicate that septins increase the bending rigidity of the lipid bilayer, further investigation is required to drive more confident conclusions due to the limitations of our experimental assay.

5.2. Outlook

Septin-membrane binding

While the importance of electrostatic forces in septin-membrane interactions becomes evident through our quantification of septin localization, it is interesting to investigate further under which conditions a septin meshwork is actually established inside the GUVs. To verify whether DOPS is simply assisting in septin recruitment and that PIP2 is specifically required for the formation of the septin membrane-bound network, FRAP experiments with GUVs containing 20% DOPS and/or 5% PIP2 could be carried out. Bleaching the GFP signal in a segment of the membrane and observing the recovered fluorescence could show whether subunit exchange takes place or if septins are stably anchored on the membrane in each case, possibly forming a rigid network. Furthermore, visualization of the structural arrangement of the membrane-bound septin on the GUV membranes could verify whether septin meshes similar to the ones observed on SLBs [33, 36] are forming as we expect. Electron microscopy could potentially be employed to image slices of resin-embedded GUVs encapsulating septin hexamers or octamers [87], allowing the investigation of the filament assembly on the inner side of the bilayer. Alternatively, septin encapsulating GUVs could be fixed, sliced, and immobilized for the filamentous network to be studied with AFM [88].

Furthermore, our understanding of the human septin-membrane binding mechanisms and of the differences between human hexamers and octamers could be improved by studies of different isoforms or mutations of septin complexes with specific domains either absent or altered. This could lead to the identification of the domains participating in septin-membrane binding and improve our understanding of the related molecular mechanisms involved. Studies of SEPT9 isoforms [74, 83] could showcase the potential importance of

SEPT9 in the difference of septin organization and function of the two oligomers [89]. Furthermore, coiled-coil truncated complexes could be investigated to assess whether and how coiled-coil spacing in a septin filament might affect meshwork formation and organization, as well as to clarify the exact role of those domains in higher order septin assembly [36].

Mechanics of a septin cortex

Due to the limitations of our experimental setup, further research is necessary to confirm septins contribution in membrane rigidification. The use of a different microscope setup that could facilitate simultaneous imaging of the lipid and the septin channel without artifacts from crosstalk would allow the classification of vesicles with and without septin localization in order to verify potential correlations between the presence of a septin meshwork and the observed increased values of the bending rigidity. Use of spinning disc confocal microscopy [90] could additionally improve the quality of the acquired data, as it would eliminate out-of focus signal from diffusing specks that significantly impedes membrane detection during the analysis [91], while allowing for low exposure times. Alternatively, phase contrast microscopy [69, 71] could eliminate the need for a membrane dye allowing us to monitor the septin distribution in the vesicle lumen, while it could potentially improve the accuracy of membrane detection as saturated signal from diffusing lipid aggregates and fluorescent signal noise would not interfere with contour detection.

However, fluctuation analysis can shed light on membrane mechanics only for lower tension regimes and thus cannot be used to assess the more rigid-looking vesicles present in our samples. As we investigate the possibility of septin-induced rigidification it is essential to determine the membrane properties in higher tension regimes and, thus, active probing methods should be pursued. Micropipette aspiration in combination with fluorescent microscopy [92, 93], would allow mechanical characterization of the GUVs. Optical interferometry-based indentation methods [94] could also be utilized as an alternative to the promising but highly challenging experimental setup of the micropipette experiments. Finally, methods such as quantitative acoustophoresis [95] could be further explored for an increased measurement throughput.

Initial trials with such approaches carried out during this project have indicated the need of further improvement of the quality of our samples. The high numbers of lipid aggregates and secondary structures present in the eDICE samples significantly impeded data acquisition and analysis as either vesicles with clean membranes were difficult to locate or out-of focus objects interfered with identifying and focusing on the GUVs, especially in setups with epifluorescence and widefield imaging combined with lower objective magnification. Thus, further optimization of the eDICE protocol should take place to reduce the number of present artifacts and ease the process of mechanical measurements. It is possible that the increased amount of lipid aggregates in the samples could relate to poor lipid solubility in the organic phase. In this case, different oil mixes could be tried out to potentially reduce the number of such artifacts. Additionally, removing the manual element of the emulsion formation for example, by vortexing, could potentially increase the robustness of GUV production. Finally, to reduce the number of artifacts found in the samples, centrifugation or a combination of filtering steps using pores of different sizes [96] could be incorporated after GUV formation, to isolate the GUVs from smaller free-floating lipid aggregates or large lipid clusters. Alternatively, other double emulsion methods could be explored for the production of GUVs encapsulating septins that might offer improved control over the prepared samples and improved suitability for mechanical measurements, such as octanol-assisted liposome

5. *Conclusion and outlook*

assembly [55], if further developed to allow for quick encapsulation of polymerizing proteins. Currently, eDICE remains the most promising method for cortex reconstitution with its experimental ease, speed and versatility in protein encapsulation.

A. Appendix

A.1. Evaluation of eDICE samples

Septin hexamers			
Sample	# of detected vesicles	# of clean vesicles	% of clean vesicles
<i>100% DOPC</i>			
A	162	67	41
B	340	129	38
C	90	9	10
D	91	6	7
<i>0% DOPS, 5% PIP2</i>			
A	969	81	8
B	2512	325	13
C	1084	3	<1
<i>20% DOPS, 0% PIP2</i>			
A	1432	53	4
B	352	1	<1
C	52	3	6
<i>20% DOPS, 1% PIP2</i>			
A	1965	191	10
B	2276	220	10
C	119	4	3
<i>20% DOPS, 3% PIP2</i>			
A	963	225	23
B	1843	32	2
C	2770	29	1
D	681	8	1
<i>20% DOPS, 5% PIP2</i>			
A	1163	291	25
B	21	2	10
C	29	2	7

Table A.1.: Overview of all different samples of GUVs encapsulating septin hexamers. The number of the detected vesicles found in DisGUVery for each sample and the corresponding number and percentages of the vesicles remaining after automatic filtering, as described in Section 2.5.3, are presented. Samples that did not contain any clean vesicles are not included and were disregarded from the analysis.

Septin octamers			
Sample	# of detected vesicles	# of clean vesicles	% of clean vesicles
100% DOPC			
A	582	92	16
B	182	3	2
C	362	27	7
0% DOPS, 5% PIP2			
A	1101	110	10
20% DOPS, 0% PIP2			
A	910	170	19
B	416	64	15
C	187	8	4
D	87	5	6
20% DOPS, 1% PIP2			
A	695	215	31
B	1126	255	23
C	170	12	7
D	63	4	6
20% DOPS, 3% PIP2			
A	705	190	27
B	810	121	15
C	793	89	11
20% DOPS, 5% PIP2			
A	613	76	12
B	421	46	11
C	402	7	2
D	85	10	12

Table A.2.: Overview of all different samples of GUVs encapsulating septin octamers. The number of the detected vesicles found in DisGUVery for each sample and the corresponding number and percentages of the vesicles remaining after automatic filtering as described in Section 2.5.3. are presented. Samples that did not contain any clean vesicles are not included and were disregarded from the analysis.

A.2. Quantification of septin localization

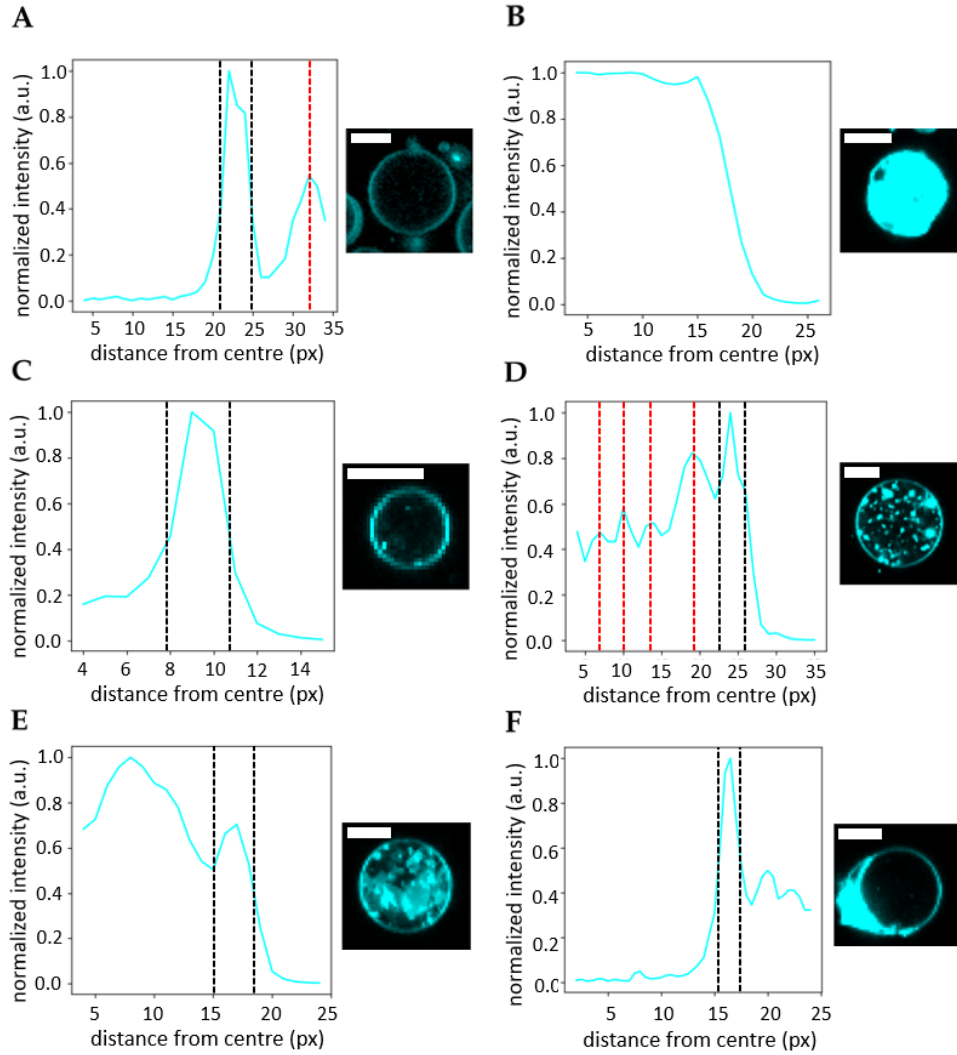


Figure A.1.: **Examples of automatic quality check of vesicle profiles.** (A) In case of additional prominent peaks in the extension of the radial profiles (red line) the tallest of the peaks close to the initial radius found in DisGUVery is selected as the position of the vesicle membrane. The borders of the estimated membrane width are indicated with black lines. (B-F) Examples of automatically rejected detections based on the conditions described in Section 2.5.3. (B) No peak was detected. (C) Wide peak detected indicating the vesicle is out of focus. (D) Multiple peaks inside the vesicle (red lines) corresponding to small lipid aggregates. (E) High lipid signal inside the vesicle indicates encapsulated large lipid aggregates. (F) High lipid signal outside the vesicle indicates attached lipid clusters. Scale bars: 5 μm .

A. Appendix

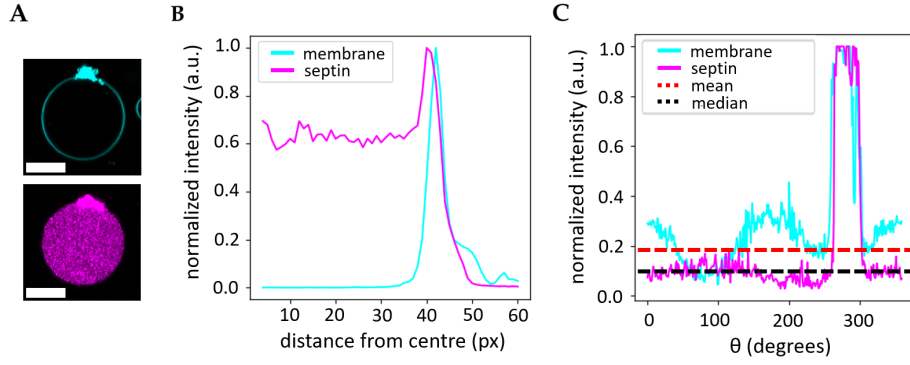


Figure A.2.: Clusters on the membrane can be misinterpreted as localization. (A) The lipid (cyan) and the septin (magenta) channel of a vesicle displaying clusters on a segment of the membrane. (B) The clusters lead to a peak in the radial intensity profile of the septin channel, even though no uniform localization is present on the membrane. (C) The cluster can be detected in the angular intensity profile of the vesicle as a deviation from the uniform signal in the majority of angles. To avoid misinterpretations, in Eq. 2.1 the median (black line) of the septin signal inside the area corresponding to the membrane width is calculated instead of the mean (red line). Scale bars: 10 μm

Septin hexamers						
PC:PS:PIP2	100:0:0	95:0:5	80:20:0	79:20:1	77:20:3	75:20:5
100:0:0	-	$1.33 \cdot 10^{-17}$	$5.62 \cdot 10^{-3}$	$1.93 \cdot 10^{-3}$	$4.35 \cdot 10^{-33}$	$3.21 \cdot 10^{-38}$
95:0:5	-	-	$5.91 \cdot 10^{-4}$	$7.96 \cdot 10^{-20}$	$1.22 \cdot 10^{-9}$	$1.06 \cdot 10^{-13}$
80:20:0	-	-	-	$7.87 \cdot 10^{-1}$	$1.55 \cdot 10^{-8}$	$3.88 \cdot 10^{-10}$
79:20:1	-	-	-	-	$4.47 \cdot 10^{-45}$	$3.38 \cdot 10^{-53}$
77:20:3	-	-	-	-	-	$2.17 \cdot 10^{-1}$
75:20:5	-	-	-	-	-	-

Septin octamers						
PC:PS:PIP2	100:0:0	95:0:5	80:20:0	79:20:1	77:20:3	75:20:5
100:0:0	-	$9.20 \cdot 10^{-5}$	$4.77 \cdot 10^{-10}$	$1.32 \cdot 10^{-25}$	$2.97 \cdot 10^{-23}$	$3.73 \cdot 10^{-23}$
95:0:5	-	-	$1.49 \cdot 10^{-2}$	$1.04 \cdot 10^{-13}$	$1.20 \cdot 10^{-11}$	$4.21 \cdot 10^{-11}$
80:20:0	-	-	-	$2.41 \cdot 10^{-14}$	$6.68 \cdot 10^{-11}$	$9.93 \cdot 10^{-9}$
79:20:1	-	-	-	-	$1.67 \cdot 10^{-1}$	$2.89 \cdot 10^{-1}$
77:20:3	-	-	-	-	-	$9.46 \cdot 10^{-1}$
75:20:5	-	-	-	-	-	-

Table A.3.: P-values of quantified septin hexamer and octamer localization (Fig. 3.9) from Mann-Whitney U test.

A.3. Vesicle Fluctuation Analysis

Acquired data						
Experiment	Sample	Total	Rejected	Analyzable	Rigid	Rigid (%)
A	control	11	6	1	4	36%
B	control	14	6	2	6	43%
C	control	16	5	5	6	37.5%
D	control	5	2	0	3	60%
A	hexamers	24	7	4	13	54%
B	hexamers	12	2	1	9	75%
C	hexamers	14	3	4	7	50%
A	octamers	20	6	6	8	40%
B	octamers	12	2	1	9	75%

Table A.4.: Numbers of all acquired time-lapse movies, and the amount of the data that was either rejected, analyzable with VFA or too rigid for analysis and the respective percentage. Videos rejected corresponded to failed detection due to artifacts, unidentifiable mode regimes, significant dependence on q_{low} choice, or small vesicles ($<4\mu\text{m}$).

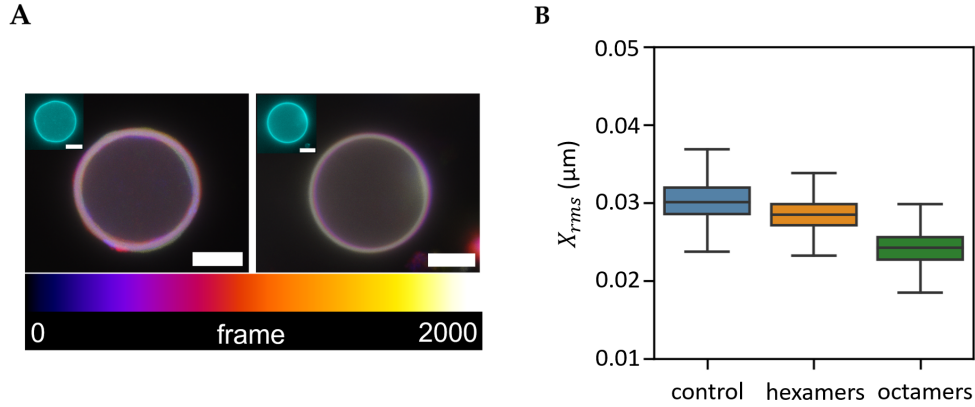


Figure A.3.: **Rigid vesicles were always present in the samples.** (A) Color coded representation of the time-lapse videos of (left) a visibly fluctuating vesicle of a DOPC control sample and (right) a rigid-looking vesicle with membrane composition 75:20:5 DOPC:DOPS:PIP2 encapsulating octamers for which no fluctuations could be detected in the script. Scale bars: $5\mu\text{m}$. (B) To quantify the magnitude of the membrane displacements of rigid-looking vesicles the distributions of the root mean square displacement of the membrane between consecutive frames, X_{rms} , was calculated for a randomly chosen vesicle found in each type of sample studied (indicated on the x-axis). Medians from left to right: 0.035 , 0.033 , $0.028\mu\text{m}$. The medians are all equal to approximately half the pixel size ($0.065\mu\text{m}$), indicating the limitations of our spatial resolution in the analysis.

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Colophon

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